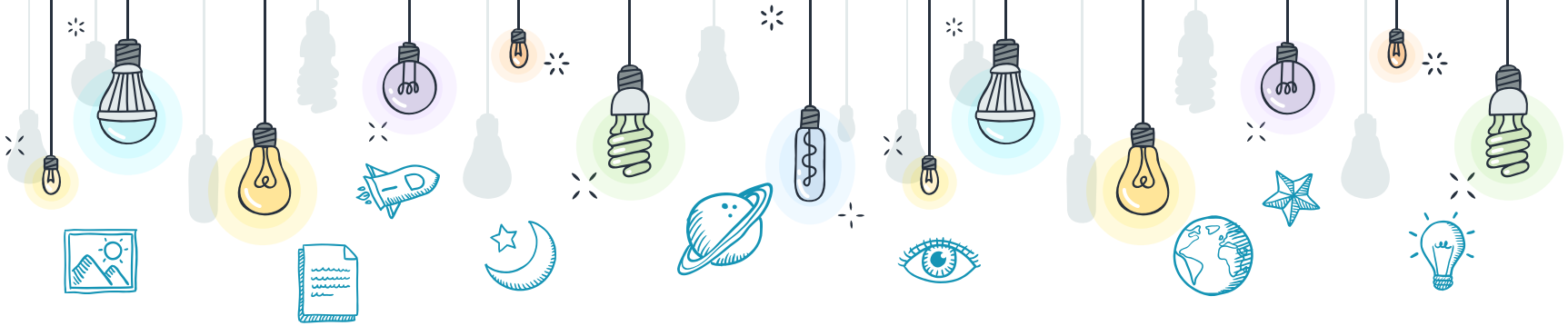


A Brief Intro to Multimodal LLM

Stay tune to MLLM tutorial series:

https://mllm2024.github.io/COLING2024

Hao Fei

Research Fellow

National University of Singapore

http://haofei.vip/

Table of Content

1 Modality

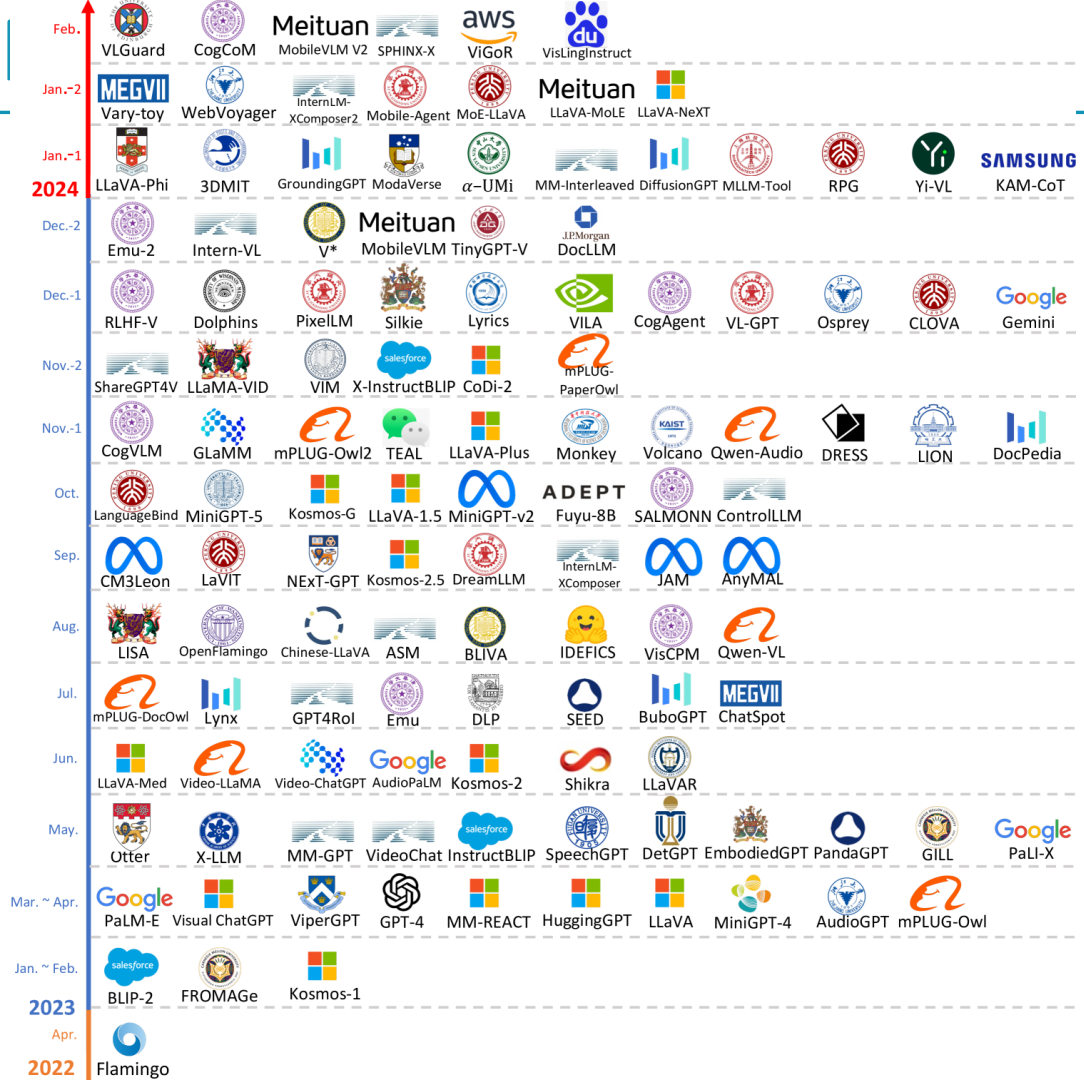
- Overview
- Multimodal Perceiving
- Multimodal Perceiving + Generation
- Unified MLLM
- Fine-grained MLLM

2 Architecture

- Overview
- Multimodal Encoding
- Tokenization
- Input-side Projection
- Backbone LLMs
- Decoding-side Connection
- Multimodal Generation

Intelligence in

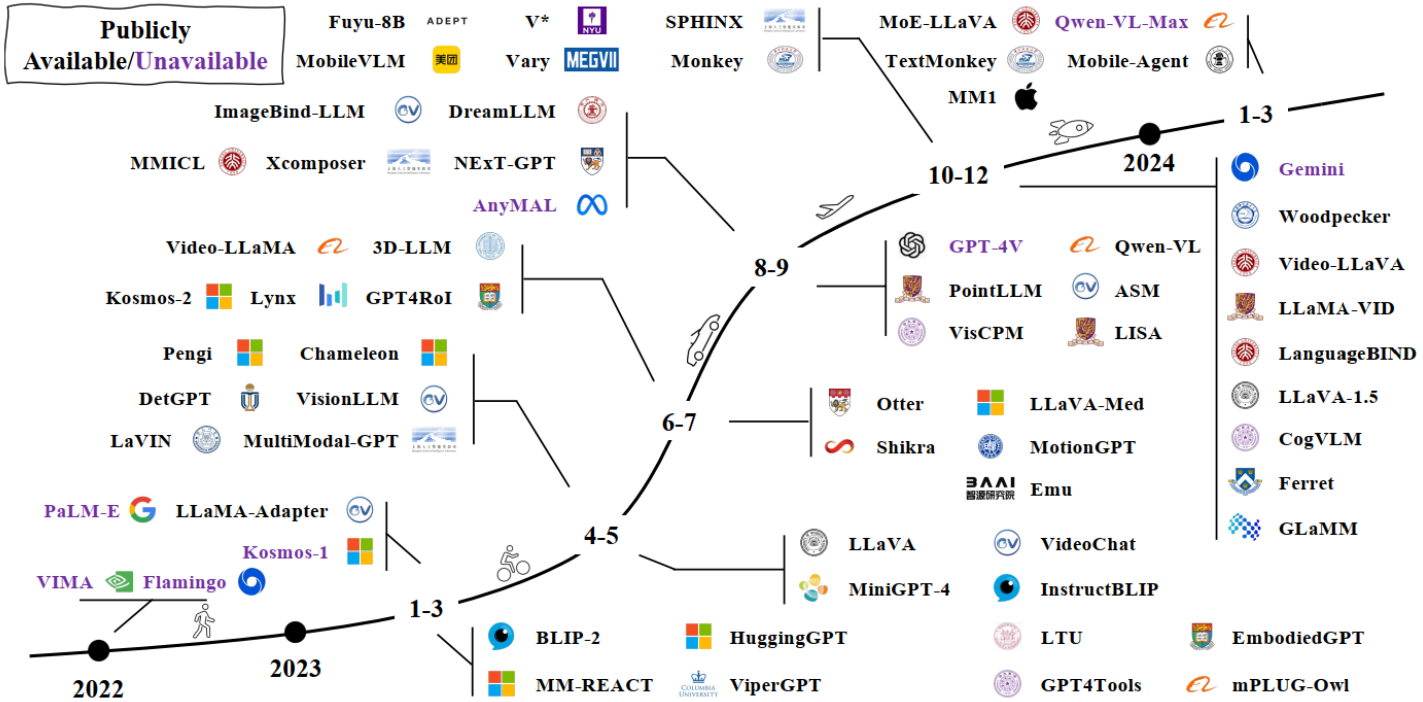
- Trends of MLLMs



[1] MM-LLMs: Recent Advances in MultiModal Large Language Models, 2023.

Intelligence in Multi-Sensory Data

- Trends of MLLMs



[1] A Survey on Multimodal Large Language Models. https://github.com/BradyFU/Awesome-Multimodal-Large-Language-Models, 2023.

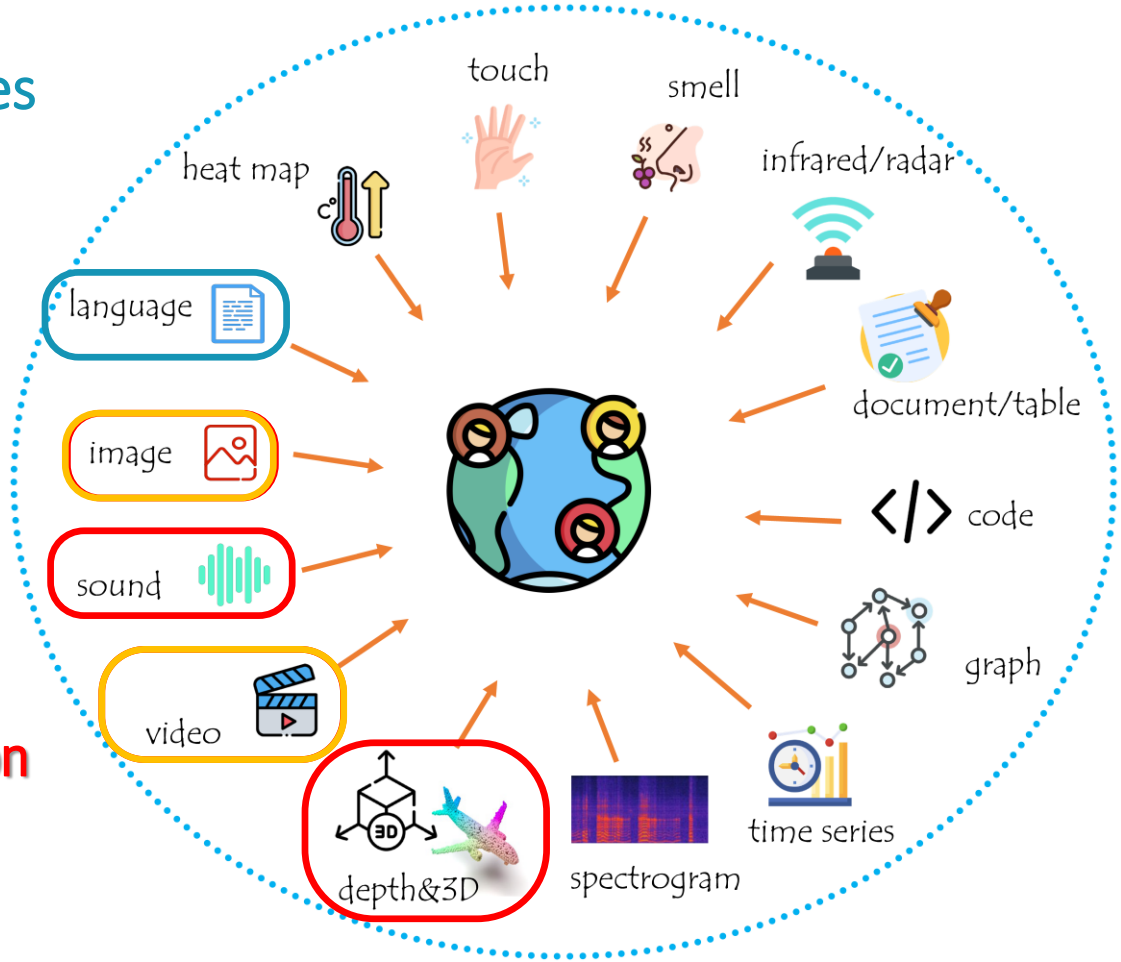
1 Modality and Functionality

What are MLLMs capable of?



Overview of Modality and Functionality

- Modalities



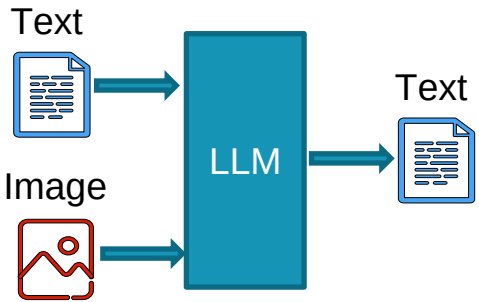
Language + **Vision**

Overview of Modality and Functionality

	Modality (w/ Language)			
	Image	**Video**	**Audio**	**3D**
Input-side Perceiving	Flamingo, Kosmos-1, Blip2, mPLUG-Owl, Mini-GPT4, LLaVA, InstructBLIP, VPGTrans, CogVLM, Monkey, Chameleon, Otter, Qwen-VL, GPT-4v, SPHINX, Yi-VL, Fuyu, …	VideoChat, Video-ChatGPT, Video-LLaMA, PandaGPT, MovieChat, Video-LLaVA, LLaMA-VID, Momentor, …	AudioGPT, SpeechGPT, VIOLA, AudioPaLM, SALMONN, MU-LLaMA, …	3D-LLM, 3D-GPT, LL3DA, SpatialVLM, PointLLM, Point-Bind, …
	[Pixel-wise] GPT4RoI, LION, MiniGPT-v2, NExT-Chat, Kosmos-2, GLaMM, LISA, DetGPT, Osprey, PixelLM, …	[Pixel-wise] PG-Video-LLaVA, Merlin, MotionEpic, …	-	-
	Video-LLaVA, Chat-UniVi, LLaMA-VID		-	-
	Panda-GPT, Video-LLaMA, AnyMAL, Macaw-LLM, Gemini, VideoPoet, ImageBind-LLM, LLMBind, LLaMA-Adapter, …			-
Perceiving + Generating	GILL, EMU, MiniGPT-5, DreamLLM, LLaVA-Plus, InternLM-XComposer2, SEED-LLaMA, LaVIT, Mini-Gemini, …	GPT4Video, Video-LaVIT, VideoPoet, …	AudioGPT, SpeechGPT, VIOLA, AudioPaLM, …	-
	[Pixel-wise] Vitron		-	-
	NExT-GPT, Unified-IO 2, AnyGPT, CoDi-2, Modaverse, ViT-Lens, …			-

Multimodal Perceiving

- Image-perceiving MLLM
  - Flamingo,
  - Kosmos-1,
  - Blip2, mPLUG-Owl,
  - Mini-GPT4, LLaVA,
  - InstructBLIP, Otter,
  - VPGTrans
  - Chameleon,
  - Qwen-VL, GPT-4v,
  - SPHINX,
  - …



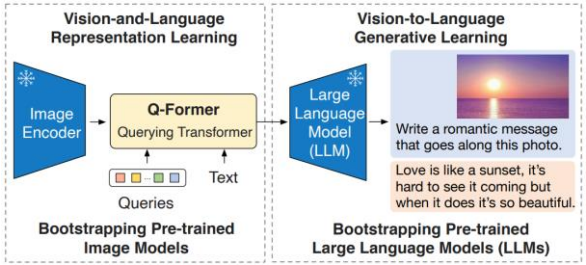
Encode input images with external image encoders, generating LLM-understandable visual feature, which is then fed into the LLM. LLM then interprets the input images based on the input text instructions and produces a textual response.

[1] Flamingo: a Visual Language Model for Few-Shot Learning. 2022
[2] Language Is Not All You Need: Aligning Perception with Language Models. 2023
[3] BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. 2023
[4] MiniGPT-4: Enhancing Vision-Language Understanding with Advanced Large Language Models. 2024
…

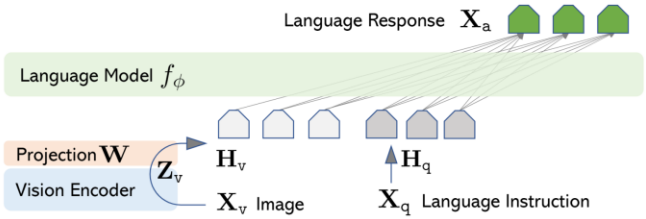
Multimodal Perceiving

- Image-perceiving MLLM

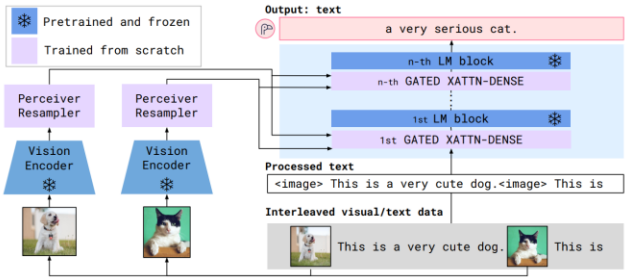
Blip2



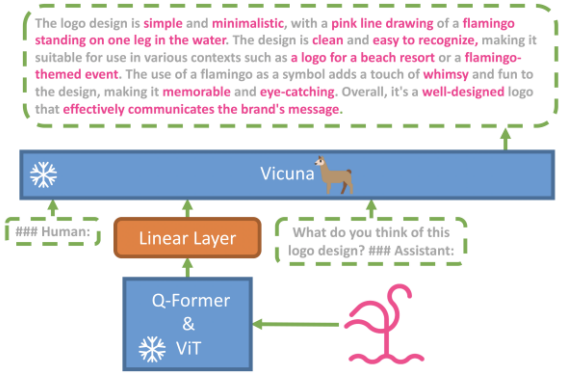
LLaVA



Flamingo



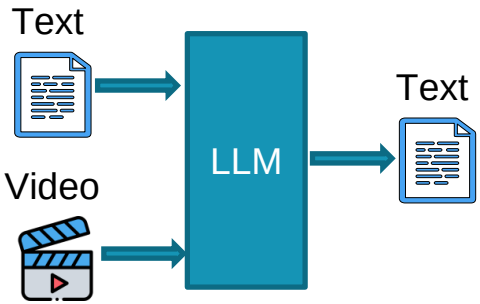
Mini-GPT4



[1] Flamingo: a Visual Language Model for Few-Shot Learning. 2022
[2] BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. 2023
[3] Visual Instruction Tuning. 2023
[4] A Survey on Multimodal Large Language Models. https://github.com/BradyFU/Awesome-Multimodal-Large-Language-Models, 2023.

Multimodal Perceiving

- ## Video-perceiving MLLM
  - VideoChat,
  - Video-ChatGPT,
  - Video-LLaMA,
  - PandaGPT,
  - MovieChat,
  - Video-LLaVA,
  - LLaMA-VID,
  - Momentor
  - …



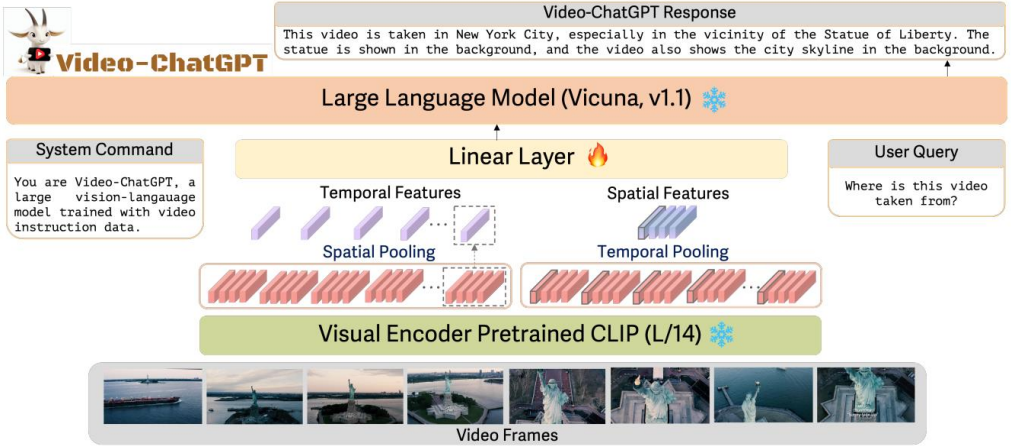
Encode input videos with external video encoders, generating LLM-understandable visual feature, feeding into LLM, which then interprets the input videos based on the input text instructions and produces a textual response.

[1] VideoChat: Chat-Centric Video Understanding. 2023
[2] Video-ChatGPT: Towards Detailed Video Understanding via Large Vision and Language Models. 2023
[3] Video-LLaMA: An Instruction-tuned Audio-Visual Language Model for Video Understanding. 2023
[4] Video-LLaVA: Learning United Visual Representation by Alignment Before Projection. 2023
[5] Momentor: Advancing Video Large Language Model with Fine-Grained Temporal Reasoning. 2024
…

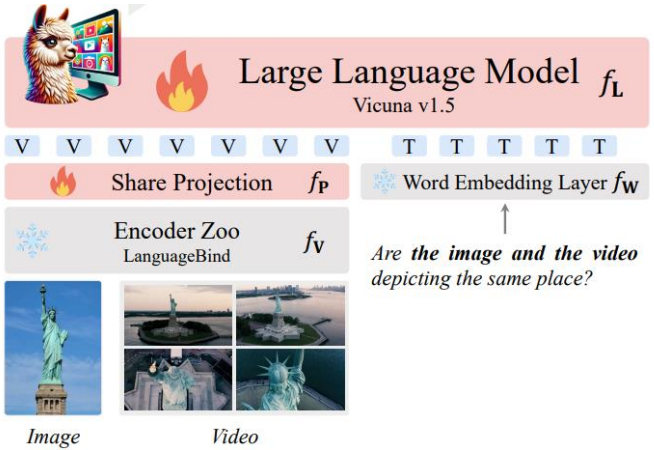
Multimodal Perceiving

- Video-perceiving MLLM

Video-ChatGPT



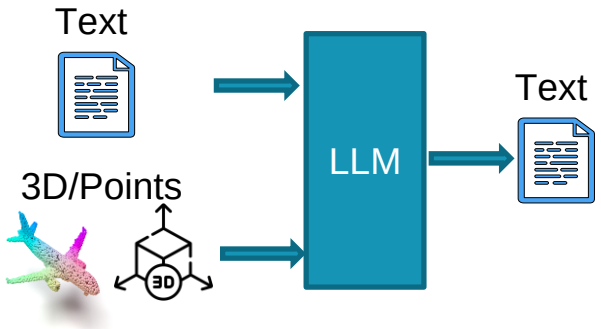
Video-LLaVA



[1] Video-ChatGPT: Towards Detailed Video Understanding via Large Vision and Language Models. 2023
[2] Video-LLaVA: Learning United Visual Representation by Alignment Before Projection. 2023
[3] Video Understanding with Large Language Models: A Survey. https://github.com/yunlong10/Awesome-LLMs-for-Video-Understanding, 2023

Multimodal Perceiving

- 3D-perceiving MLLM
  - 3D-LLM,
  - 3D-GPT,
  - LL3DA,
  - SpatialVLM
  - PointLLM
  - Point-Bind
  - …

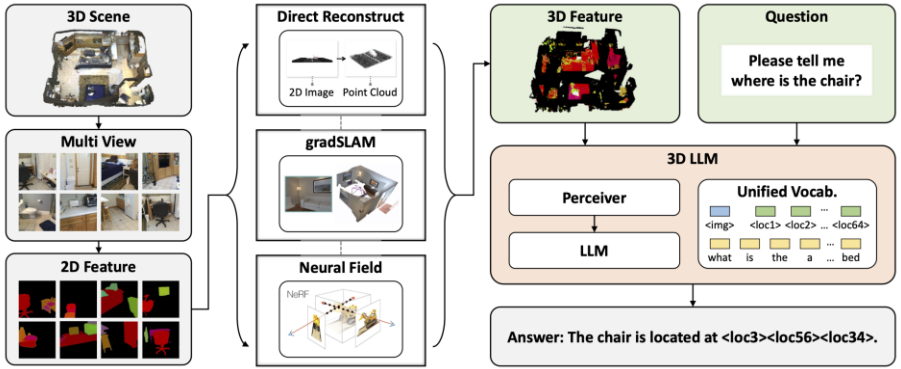


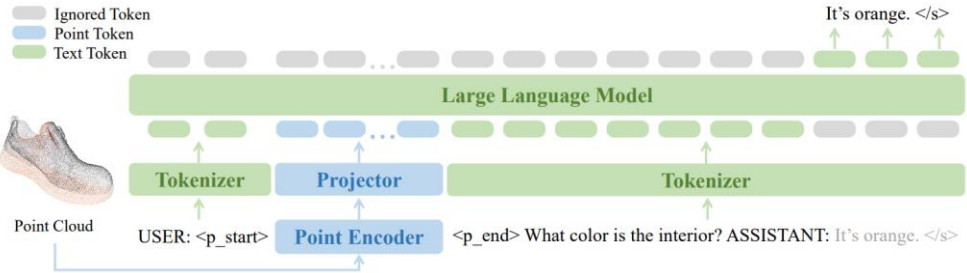
Encode input 3D information with external encoders, generating LLM-understandable 3D feature, feeding into LLM, which then interprets the input 3D/points based on the input text instructions and produces a textual response.

[1] 3D-LLM: Injecting the 3D World into Large Language Models. 2023
[2] 3D-GPT: Procedural 3D Modeling with Large Language Models. 2023
[3] LL3DA: Visual Interactive Instruction Tuning for Omni-3D Understanding, Reasoning, and Planning. 2023
[4] PointLLM: Empowering Large Language Models to Understand Point Clouds. 2023
[5] SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. 2024
…

Multimodal Perceiving

- 3D-perceiving MLLM
  - 3D-LLM
  - PointLLM



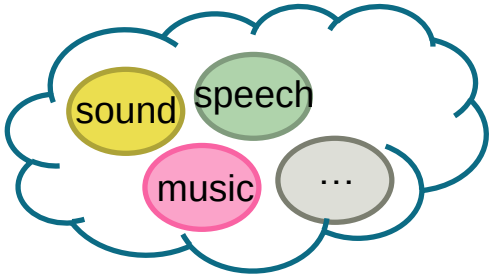


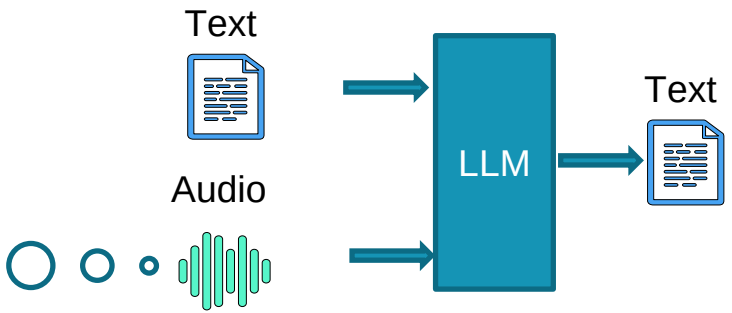
[1] 3D-LLM: Injecting the 3D World into Large Language Models. 2023
[2] PointLLM: Empowering Large Language Models to Understand Point Clouds. 2023

Multimodal Perceiving

- ## Audio-perceiving MLLM

  - AudioGPT,
  - SpeechGPT,
  - VIOLA,
  - AudioPaLM
  - SALMONN
  - MU-LLaMA
  - …





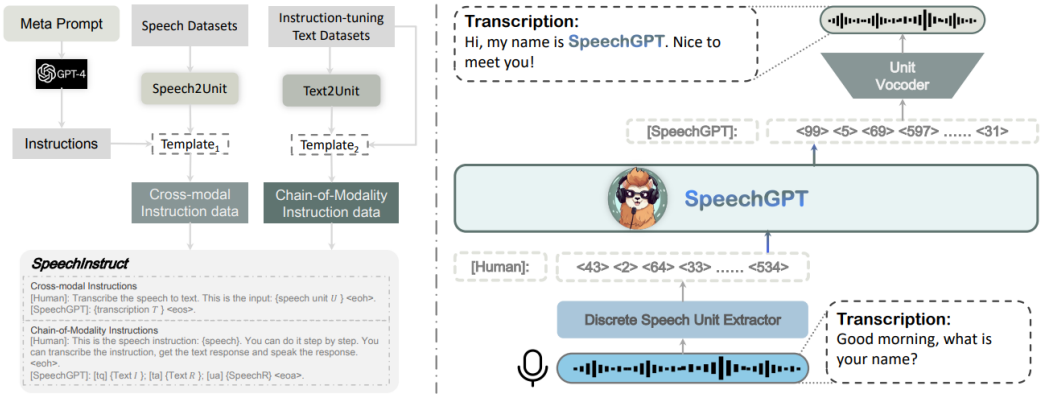
Encode input audio signals with external encoders, generating LLM-understandable signal features, feeding into LLM, which then interprets the audio based on the input text instructions and produces a textual response.

[1] AudioGPT: Understanding and Generating Speech, Music, Sound, and Talking Head. 2023
[2] SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities. 2023
[3] VioLA: Unified Codec Language Models for Speech Recognition, Synthesis, and Translation. 2023
[4] AudioPaLM: A Large Language Model That Can Speak and Listen. 2023
[5] SALMONN: Towards Generic Hearing Abilities for Large Language Models. 2023
…

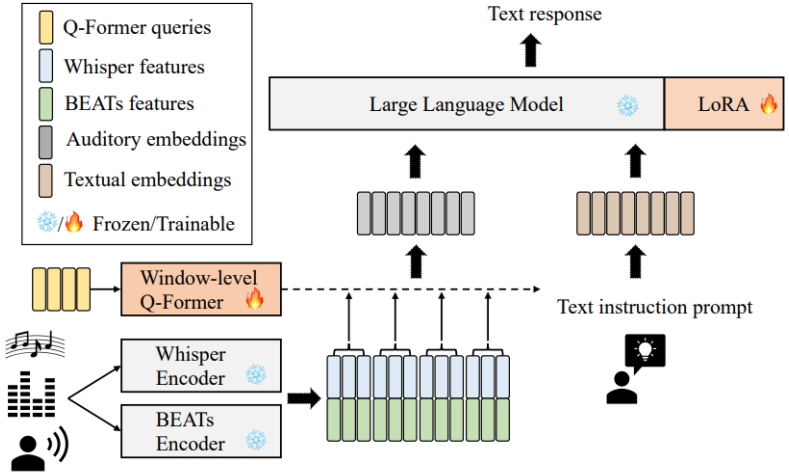
Multimodal Perceiving

- Audio-perceiving MLLM

SpeechGPT



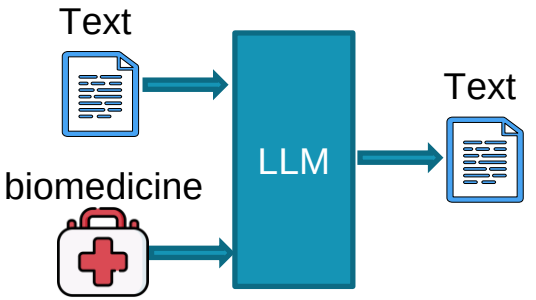
SALMONN



[1] SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities. 2023
[2] SALMONN: Towards Generic Hearing Abilities for Large Language Models. 2023
[3] Sparks of Large Audio Models: A Survey and Outlook. https://github.com/EmulationAI/awesome-large-audio-models, 2023

Multimodal Perceiving

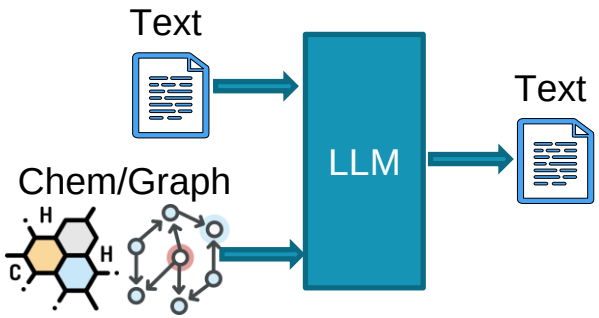
- X-perceiving MLLM
  - Bio-/Medical & Healthcare
    - BioGPT
    - DrugGPT
    - BioMedLM
    - OphGLM
    - GatorTron
    - GatorTronGPT
    - MEDITRON
    - DoctorGLM
    - BianQue
    - ClinicalGPT
    - Qilin-Med
    - ChatDoctor
    - BenTsao
    - HuatuoGPT
    - MedAlpaca
    - AlpaCare
    - Zhongjing
    - PMC-LLaMA
    - CPLLM
    - MedPaLM 2
    - BioMedGPT



[1] BioGPT: Generative Pre-trained Transformer for Biomedical Text Generation and Mining. 2022
[2] DrugGPT: A GPT-based Strategy for Designing Potential Ligands Targeting Specific Proteins. 2023
[3] MEDITRON-70B: Scaling Medical Pretraining for Large Language Models. 2023
[4] HuaTuo: Tuning LLaMA Model with Chinese Medical Knowledge. 2023
[5] AlpaCare:Instruction-tuned Large Language Models for Medical Application. 2023
[6] A Survey of Large Language Models in Medicine: Progress, Application, and Challenge, https://github.com/AI-in-Health/MedLLMsPracticalGuide. 2023.
…

Multimodal Perceiving

- X-perceiving MLLM
  - Molecule & Chemistry
    - ChemGPT
    - SPT
    - T5 Chem
    - ChemLLM
    - MolCA
    - MolXPT
    - MolSTM
    - GIMLET
    - …
  - Graph
    - StructGPT
    - GPT4Graph
    - GraphGPT
    - LLaGA
    - HiGPT
    - …
  - Geographical Information System (GIS)
    - GeoGPT



[1] Neural Scaling of Deep Chemical Models. 2022
[2] ChemLLM: A Chemical Large Language Model. 2023
[3] MolCA: Molecular Graph-Language Modeling with Cross-Modal Projector and Uni-Modal Adapter. 2023
[4] StructGPT: A General Framework for Large Language Model to Reason on Structured Data. 2023
[5] LLaGA: Large Language and Graph Assistant. 2023
[6] Awesome-Graph-LLM, https://github.com/XiaoxinHe/Awesome-Graph-LLM. 2023

Unified MLLM: Perceiving + Generation

- ## Scenarios

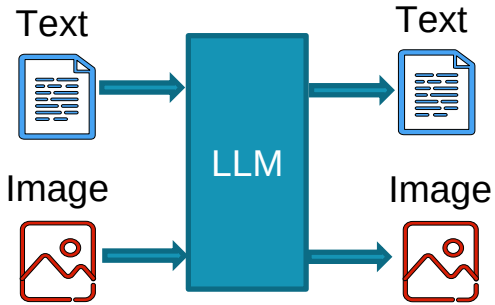


*Often, MLLMs need to not only **understand** the input multimodal information, but also to **generate** information in that modality.*

- Image Captioning
- Visual Question Answering
- Text-to-Vision Synthesis
- Vision-to-Vision Translation
- Scene Text Recognition
- Scene Text Inpainting
- …

Unified MLLM: Perceiving + Generation

- Image
  - GILL
  - EMU
  - MiniGPT-5
  - DreamLLM
  - LLaVA-Plus
  - LaVIT
  - …

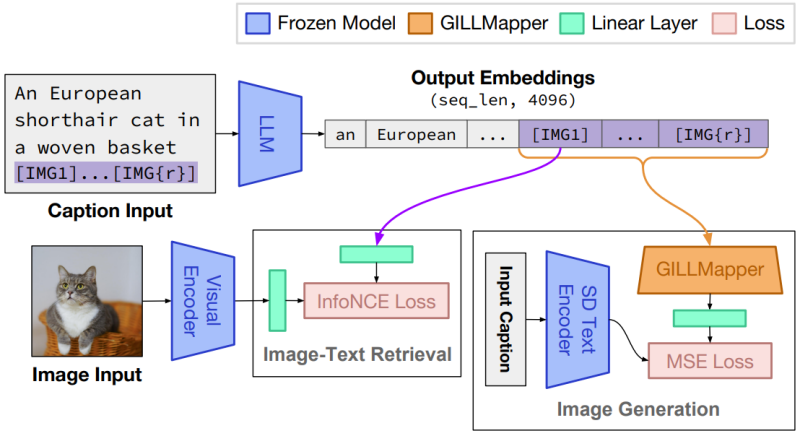


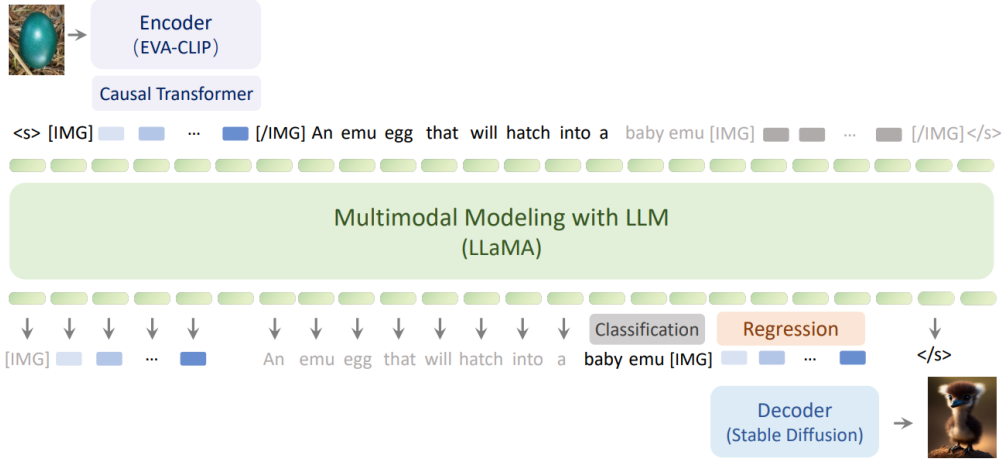
Central LLMs take as input both texts and images, after semantics comprehension, and generate both texts and images.

[1] Generating Images with Multimodal Language Models. 2023
[2] Generative Pretraining in Multimodality. 2023
[3] MiniGPT-5: Interleaved Vision-and-Language Generation via Generative Vokens. 2023
[4] DreamLLM: Synergistic Multimodal Comprehension and Creation. 2023
[5] LLaVA-Plus: Learning to Use Tools for Creating Multimodal Agents. 2023
…

Unified MLLM: Perceiving + Generation

- Image
  - GILL
  - EMU

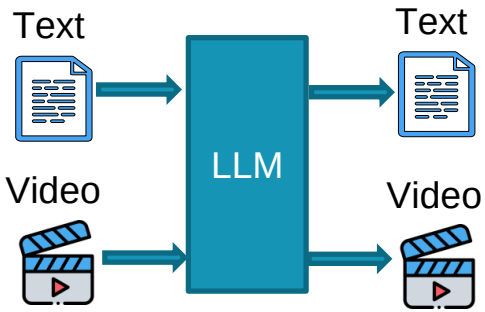




[1] Generating Images with Multimodal Language Models. 2023
[2] Generative Pretraining in Multimodality. 2023

Unified MLLM: Perceiving + Generation

- Video
  - GPT4Video
  - VideoPoet
  - Video-LaVIT
  - …



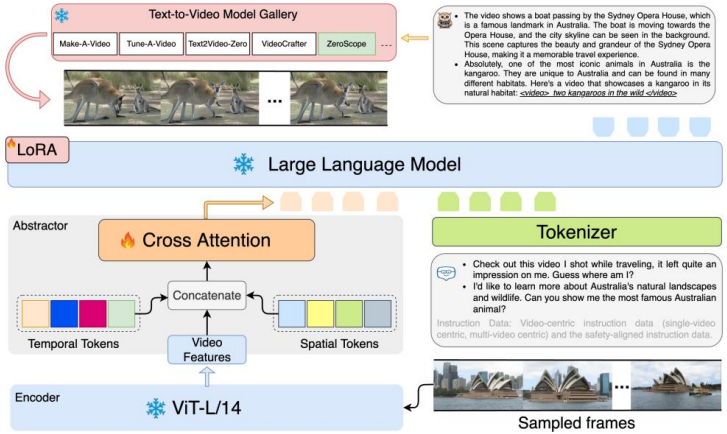
Central LLMs take as input both texts and videos, after semantics comprehension, and generate both texts and videos.

[1] GPT4Video: A Unified Multimodal Large Language Model for Instruction-Followed Understanding and Safety-Aware Generation. 2023
[2] VideoPoet: A Large Language Model for Zero-Shot Video Generation. 2023
[3] Video-LaVIT: Unified Video-Language Pre-training with Decoupled Visual-Motional Tokenization. 2024
…

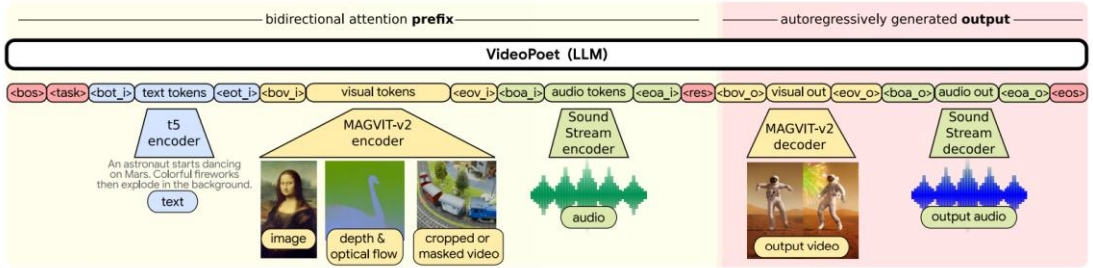
Unified MLLM: Perceiving + Generation

- Video

GPT4Video



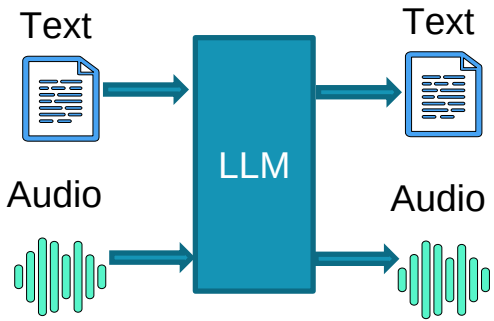
VideoPoet



[1] GPT4Video: A Unified Multimodal Large Language Model for Instruction-Followed Understanding and Safety-Aware Generation. 2023
[2] VideoPoet: A Large Language Model for Zero-Shot Video Generation. 2023

Unified MLLM: Perceiving + Generation

- Audio
  - AudioGPT,
  - SpeechGPT,
  - VIOLA,
  - AudioPaLM,
  - …



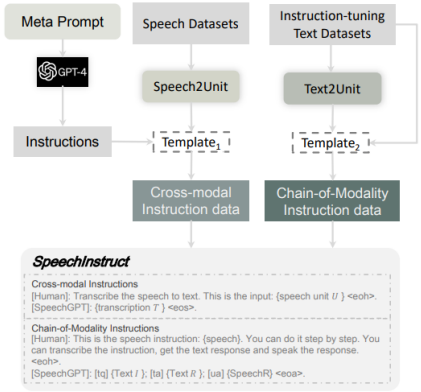
Central LLMs take as input both texts and audio, after semantics comprehension, and generate both texts and audio.

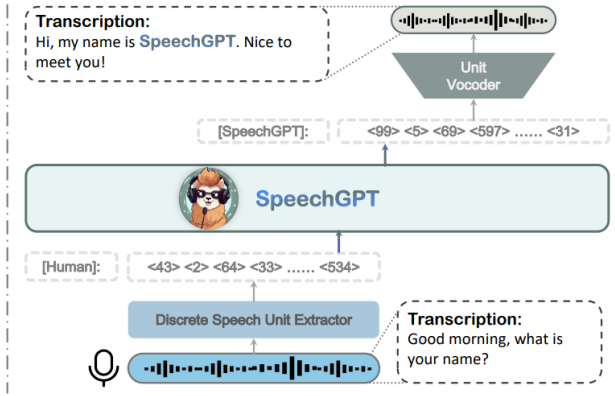
[1] AudioGPT: Understanding and Generating Speech, Music, Sound, and Talking Head. 2023
[2] SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities. 2023
[3] VioLA: Unified Codec Language Models for Speech Recognition, Synthesis, and Translation. 2023
[4] AudioPaLM: A Large Language Model That Can Speak and Listen. 2023
…

Unified MLLM: Perceiving + Generation

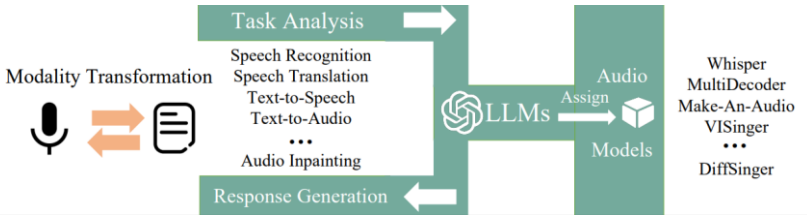
- Audio

SpeechGPT





AudioGPT



[1] SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities. 2023
[2] AudioGPT: Understanding and Generating Speech, Music, Sound, and Talking Head. 2023

Unified MLLM: Harnessing Multi-Modalities

- **Scenarios:**

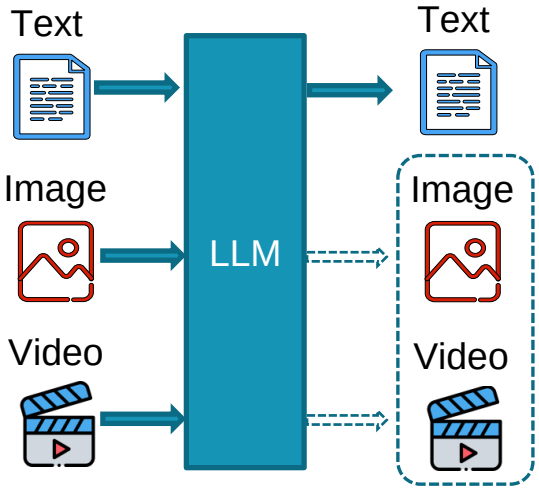


*In reality, modalities often have strong interconnections simultaneously. Thus, it is frequently necessary for MLLMs to handle the understanding of **multiple non-textual modalities at once**, rather than just one single (non-textual) modality.*

- Image+Video
- Audio+Video
- Image+Video+Audio
- Any-to-Any
- …

Unified MLLM: Harnessing Multi-Modalities

- Text+Image+Video
  - Video-LLaVA
  - Chat-UniVi
  - LLaMA-VID
  - …

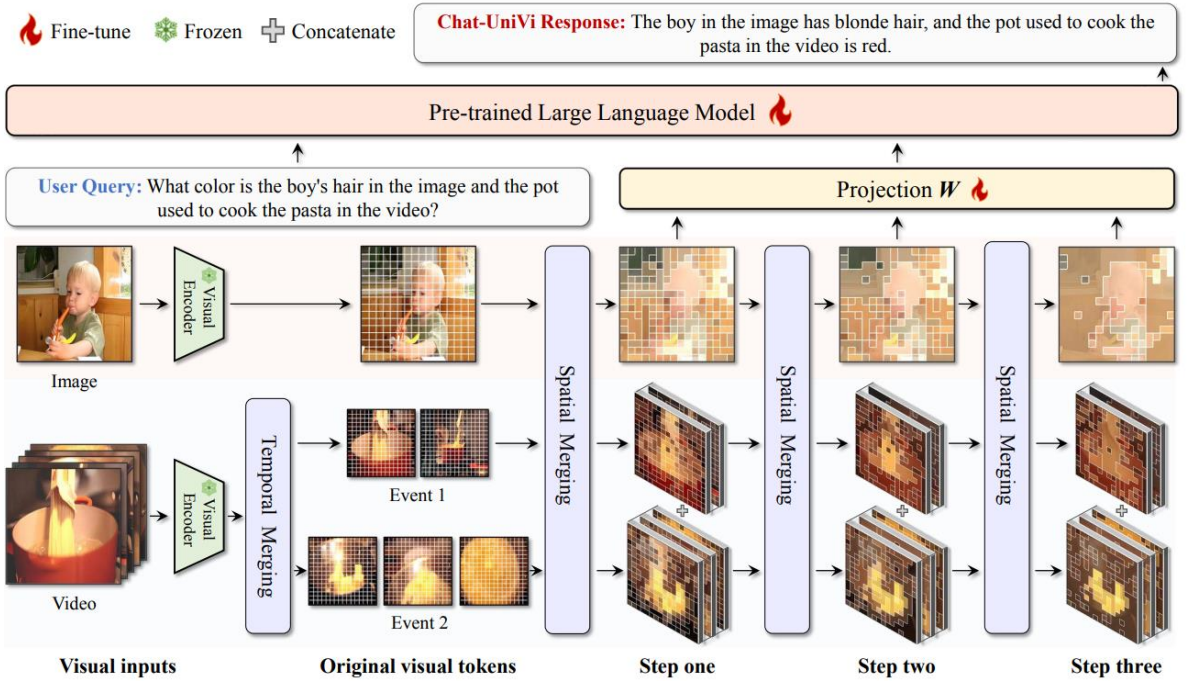


Central LLMs take as input texts, image and video, after semantics comprehension, and generate texts (maybe also image and video, or combination).

[1] Video-LLaVA: Learning United Visual Representation by Alignment Before Projection. 2023
[2] Chat-UniVi: Unified Visual Representation Empowers Large Language Models with Image and Video Understanding. 2023
[3] LLaMA-VID: An Image is Worth 2 Tokens in Large Language Models. 2023
…

Unified MLLM: Harnessing Multi-Modalities

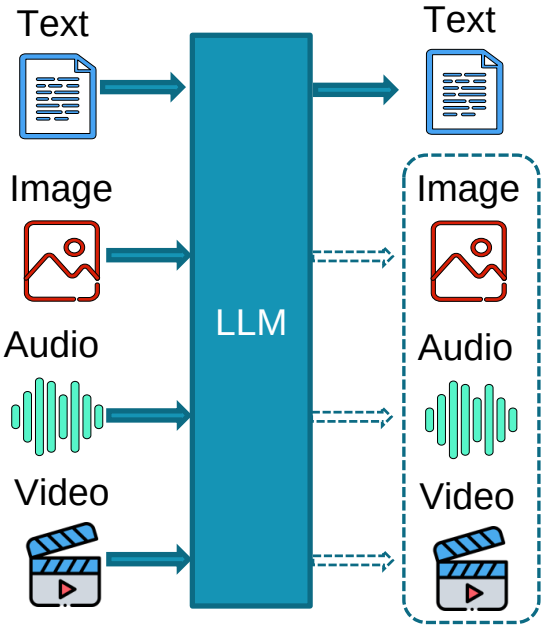
- Text+Image+Video
  - Chat-UniVi



[1] Chat-UniVi: Unified Visual Representation Empowers Large Language Models with Image and Video Understanding. 2023

Unified MLLM: Harnessing Multi-Modalities

- Text+Image+Video+Audio
  - Panda-GPT
  - Video-LLaMA
  - AnyMAL
  - Macaw-LLM
  - VideoPoet
  - ImageBind-LLM
  - LLMBind
  - LLaMA-Adapter
  - …

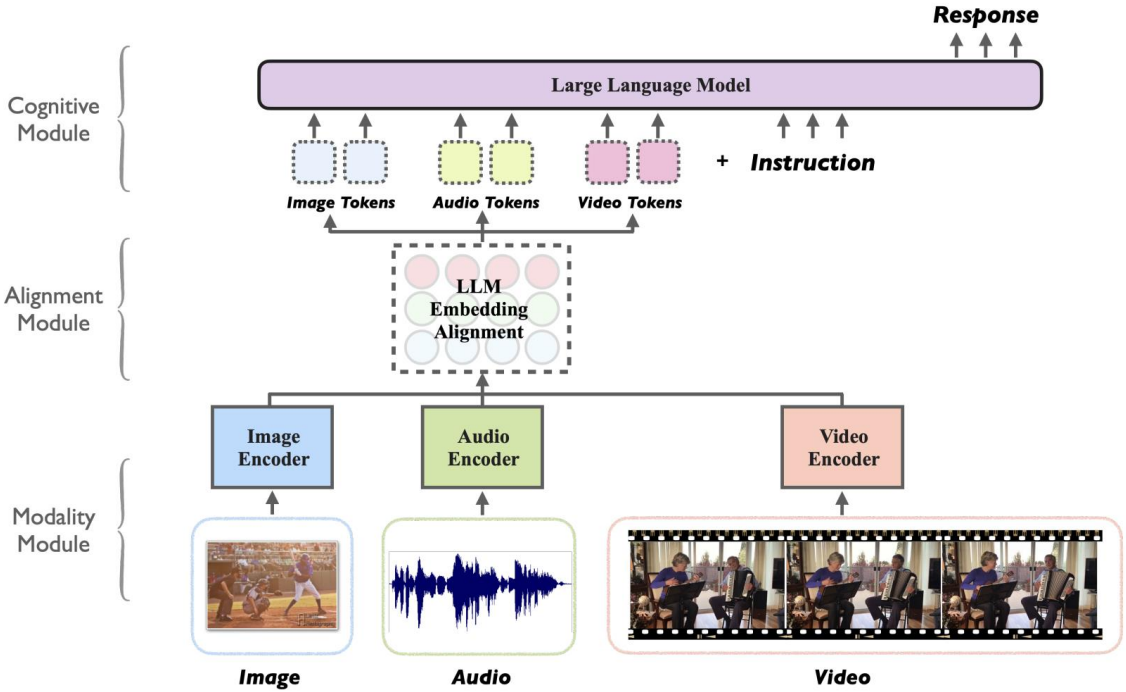


Central LLMs take as input texts, audio, image and video, and generate texts (maybe also audio, image and video, or combination).

[1] PandaGPT: One Model to Instruction-Follow Them All. 2023
[2] Video-LLaMA: An Instruction-tuned Audio-Visual Language Model for Video Understanding. 2023
[3] AnyMAL: An Efficient and Scalable Any-Modality Augmented Language Model. 2023
[4] Macaw-LLM: Multi-Modal Language Modeling with Image, Audio, Video, and Text Integration. 2023
…

Unified MLLM: Harnessing Multi-Modalities

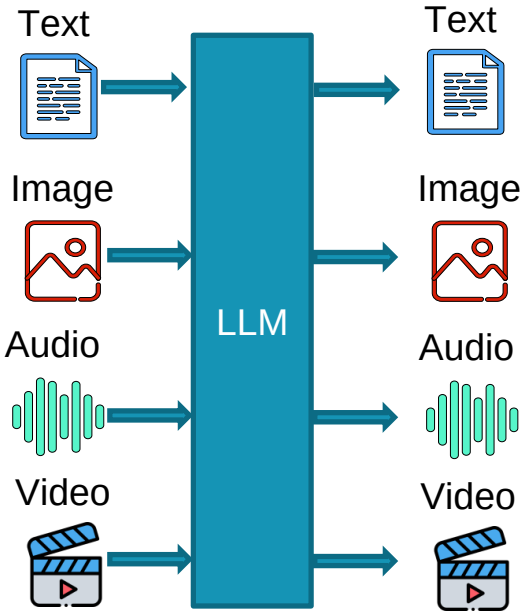
- Text+Image+Video+Audio
  - Macaw-LLM



[1] Macaw-LLM: Multi-Modal Language Modeling with Image, Audio, Video, and Text Integration. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT
  - Unified-IO 2 (w/o video)
  - AnyGPT (w/o video)
  - CoDi-2
  - Modaverse
  - …



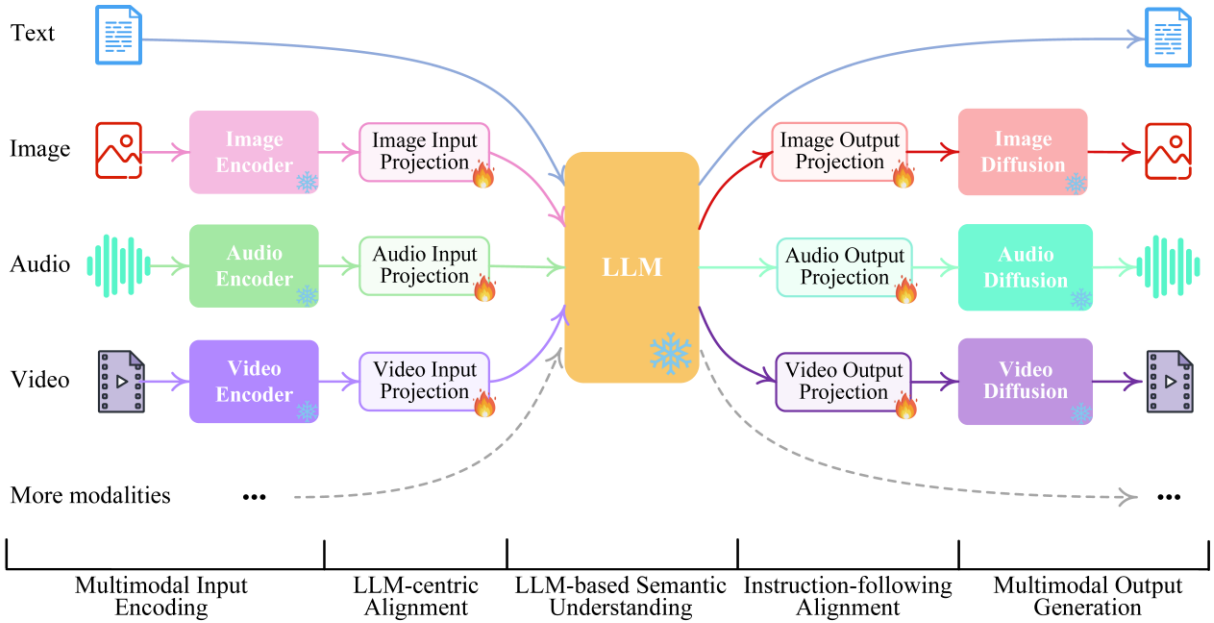
Central LLMs take as input texts, audio, image and video, and freely generate texts, audio, image and video, or combination.

[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023
[2] AnyGPT: Unified Multimodal LLM with Discrete Sequence Modeling. 2023
[3] CoDi-2: In-Context, Interleaved, and Interactive Any-to-Any Generation. 2023
[4] ModaVerse: Efficiently Transforming Modalities with LLMs. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT





Project: https://next-gpt.github.io

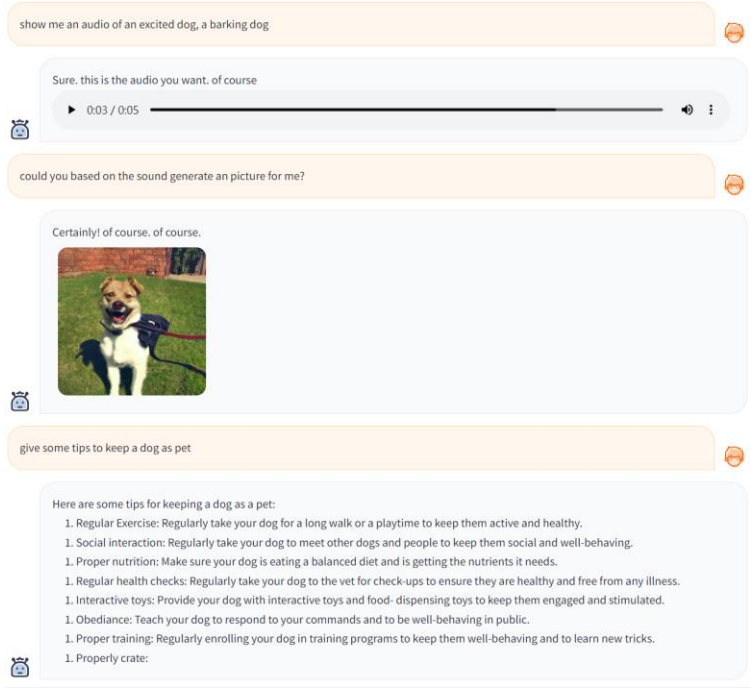
Paper: https://arxiv.org/pdf/2309.05519

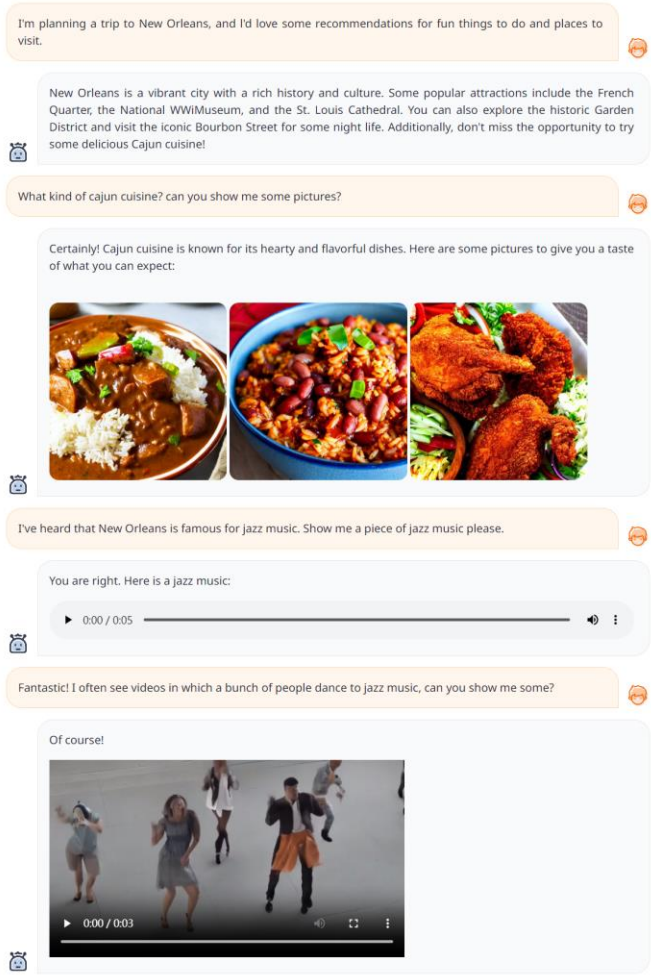
Code: https://github.com/NExT-GPT/NExT-GPT

[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT





[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT

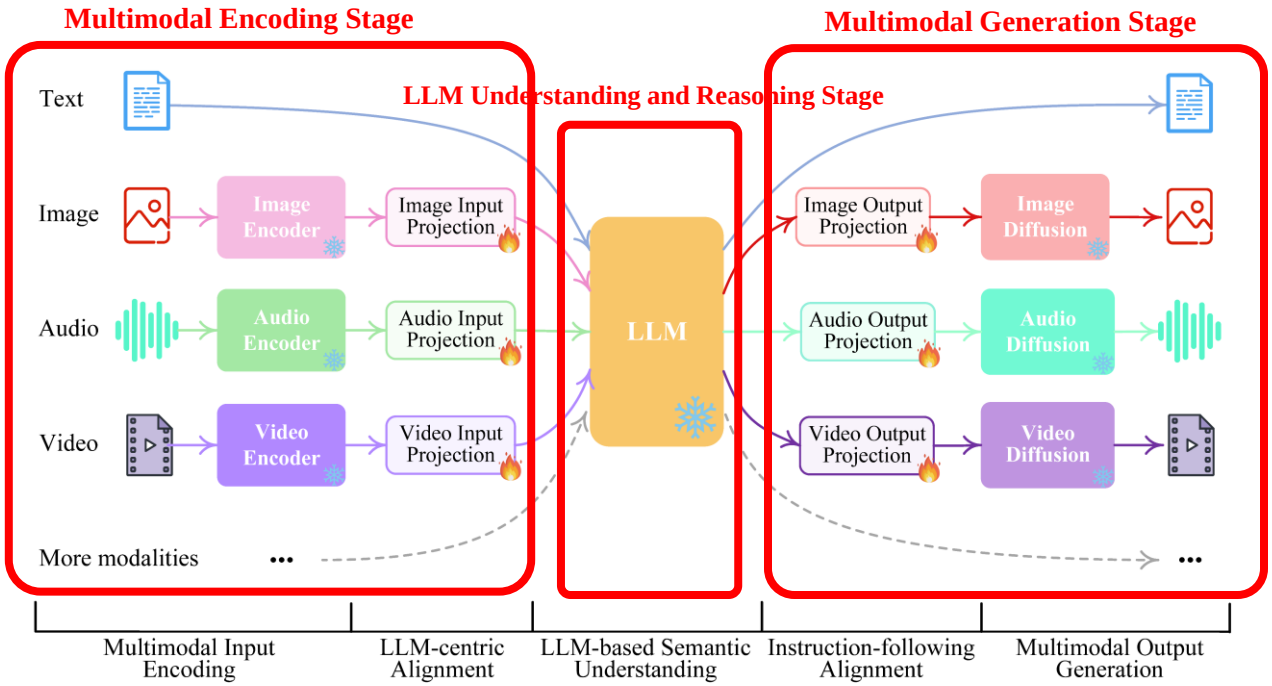




[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT

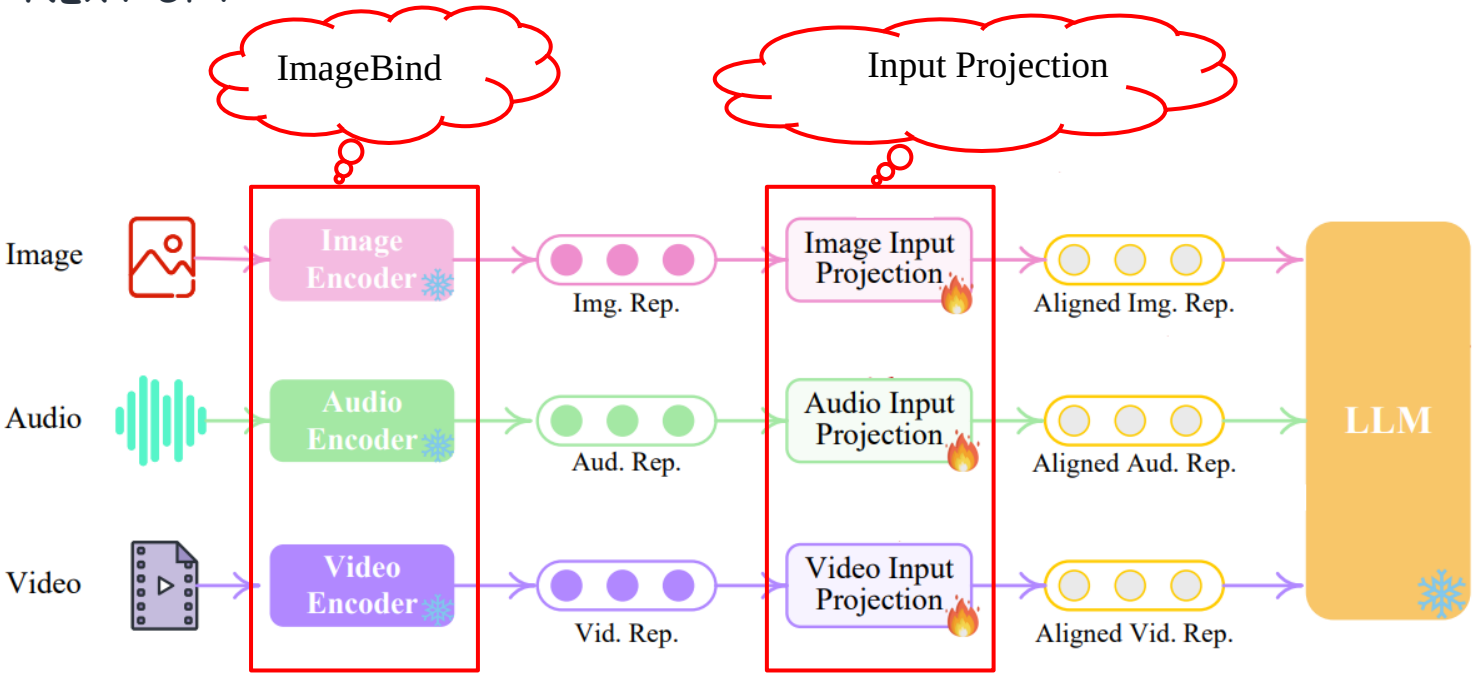


[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT

- Taking ImageBind as the unified multimodal encoder
- An input projection layer to connect multimodal encoder and LLM

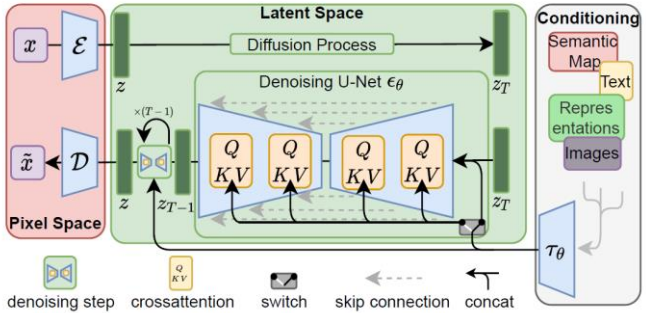


[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

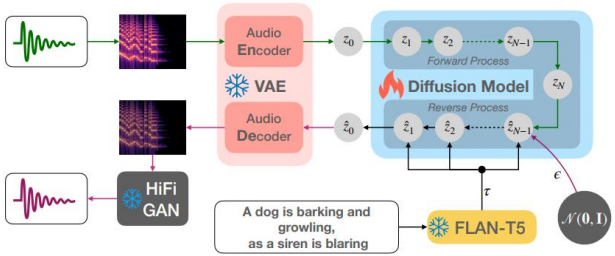
- ## Any-to-Any MLLM

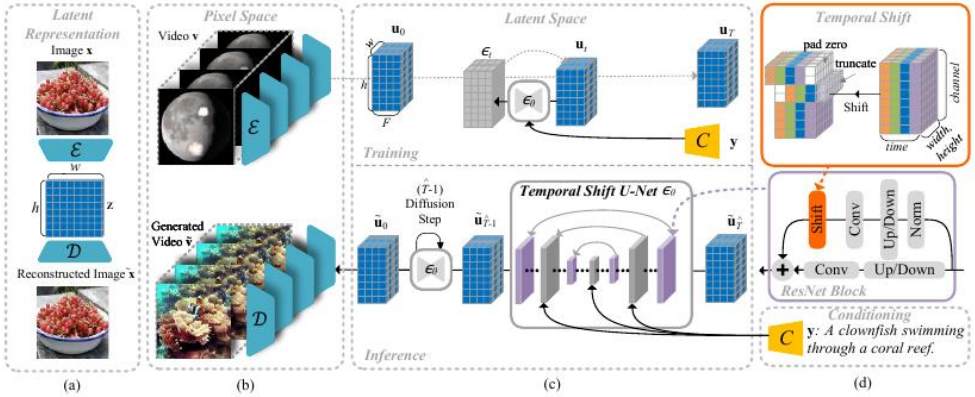
  - ### NExT-GPT



➢ Leveraging the current SoTA diffusion-based Text-to-Image, Video, Audio generation model to generate multimodal content

- Text encoder – control the generation process
- VAE
- UNet

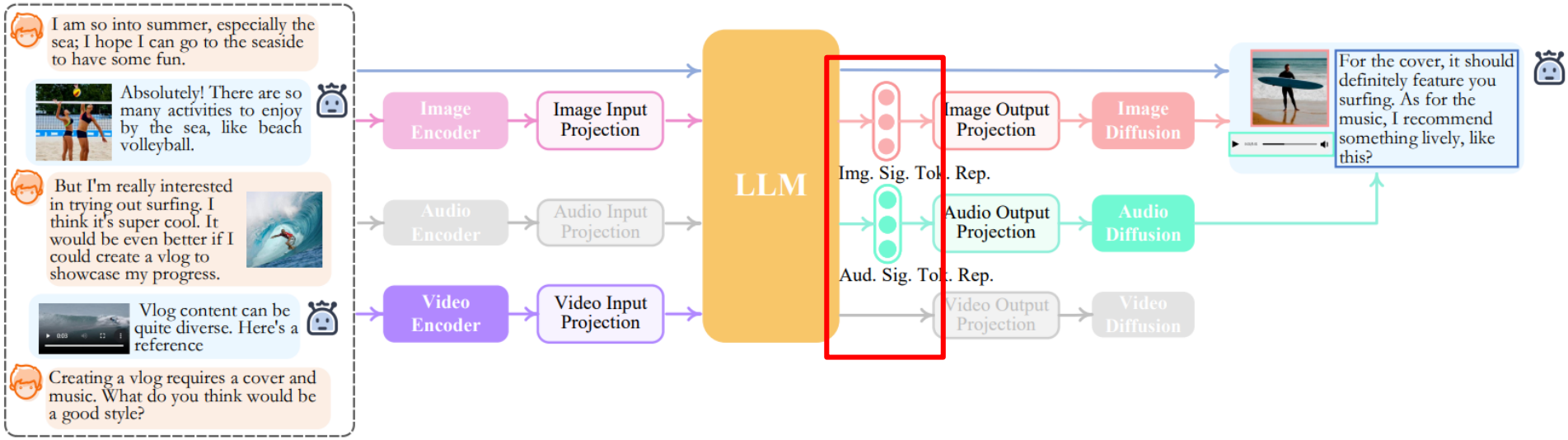




Unified MLLM: Harnessing Multi-Modalities

- **Any-to-Any MLLM**
  - NExT-GPT

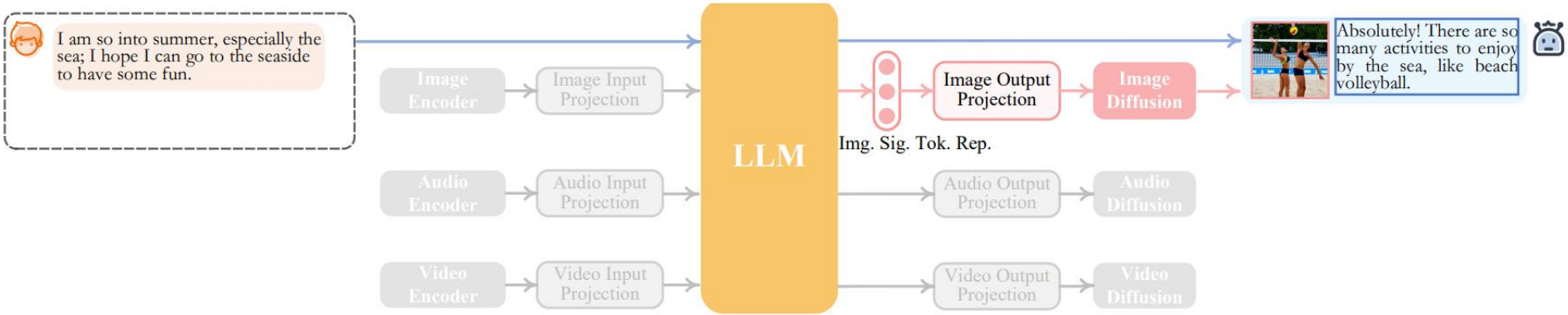
- Harnessing LLM as the core to decide whether & what modal content to output correspondingly.
- Instead of generating textual instructions, LLM produces unique "modality signal" tokens that serve as instructions to guide the generation process.



[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

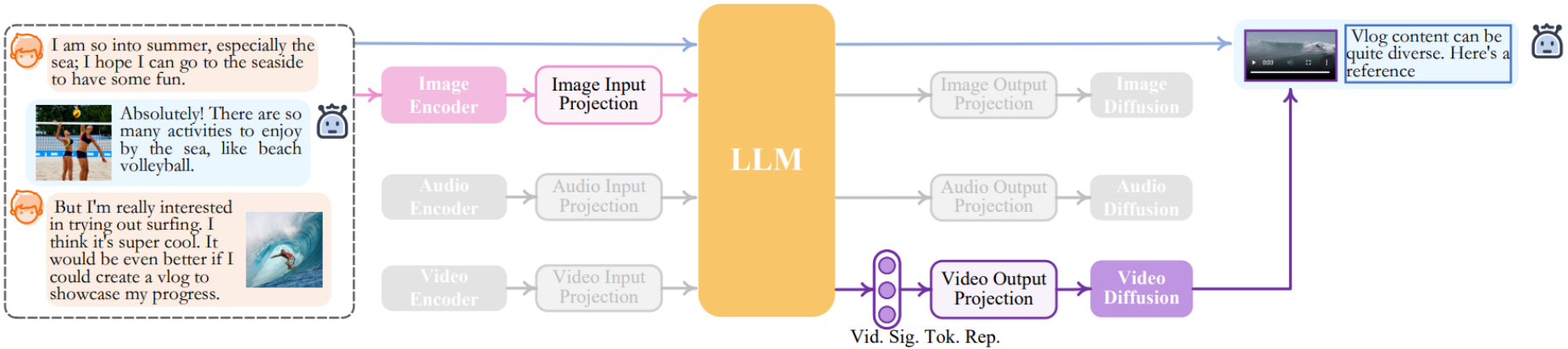
- Any-to-Any MLLM
  - NExT-GPT



[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

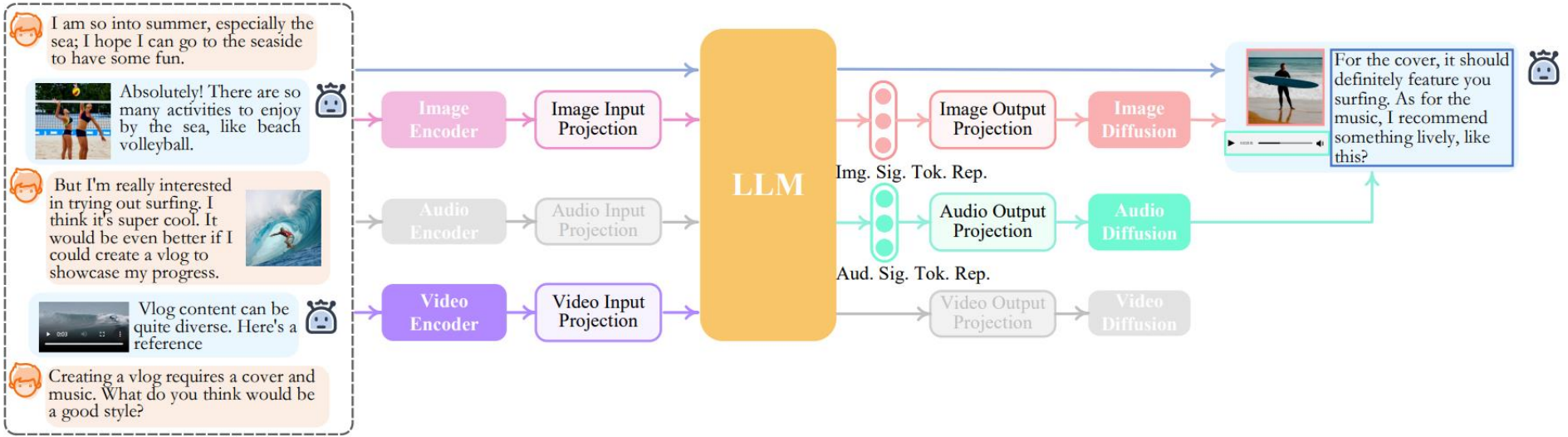
- Any-to-Any MLLM
  - NExT-GPT



[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

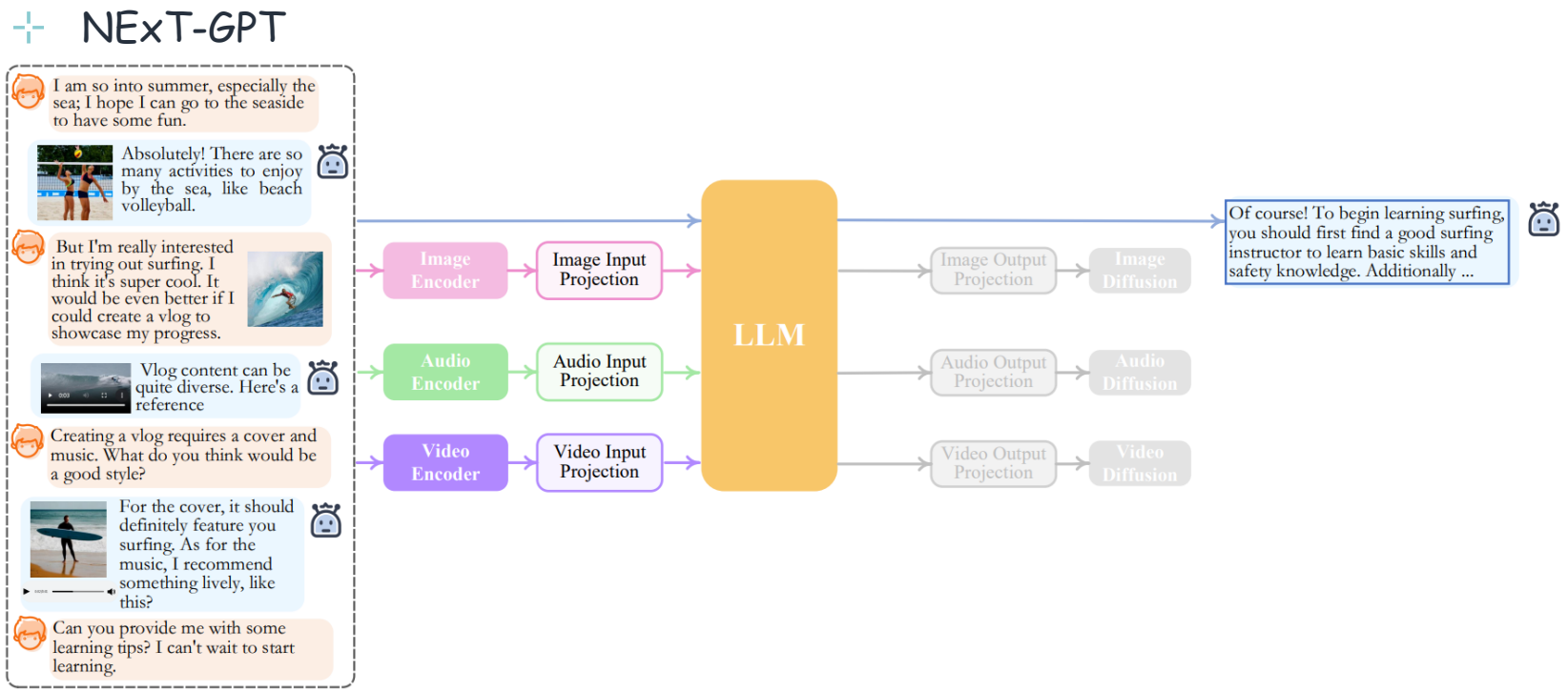
- Any-to-Any MLLM
  - NExT-GPT



[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM



[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT
    - ➢ Key Aspect-I: Parameter-efficient Low-cost Training

	Encoder		Input Projection		LLM		Output Projection		Diffusion	
	Name	Param	Name	Param	Name	Param	Name	Param	Name	Param
Text	—	—	—	—			—	—	—	—
Image					Vicuna [9]	7B❄	Transformer	31M🔥	SD [43]	1.3B❄
Audio	ImageBind [15]	1.2B❄	Linear	4M🔥	(LoRA	33M🔥)	Transformer	31M🔥	AudioLDM [34]	975M❄
Video							Transformer	32M🔥	Zeroscope [5]	1.8B❄

Table 1: Summary of system configuration. Only 1% parameters need updating.

$$\frac{Tuned\ Params}{Frozen + Tuned\ Params} = \frac{(4M+33M+31M+31M+32M)}{(4M+33M+31M+31M+32M) + (1.2B+7B+1.3B+1.8B+0.975B)}$$

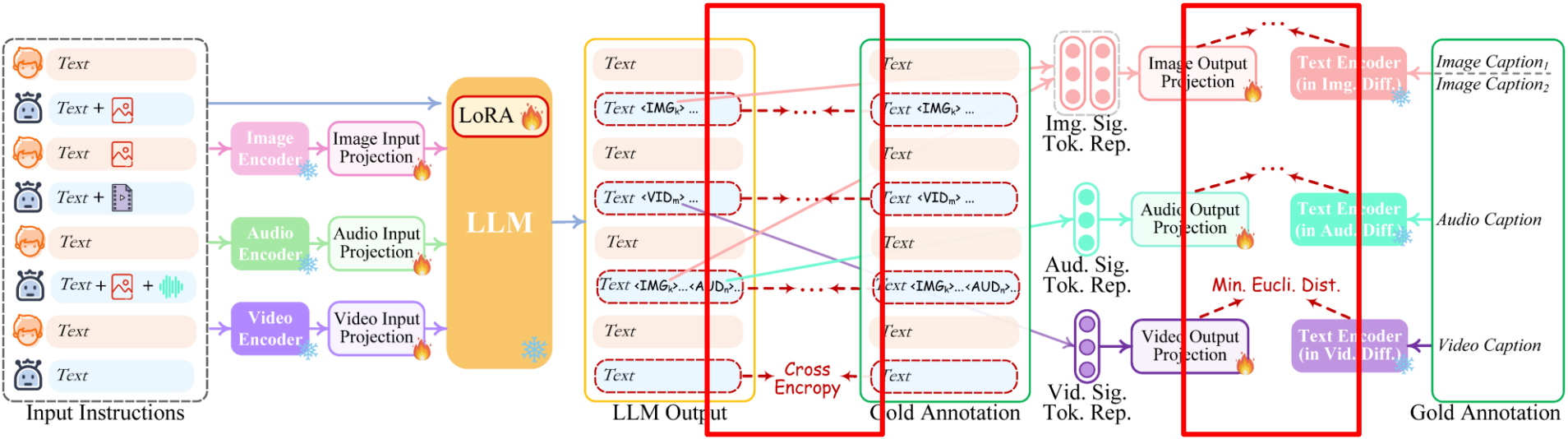
$$= \frac{131M}{131M + 12.275B} \cong \mathbf{0.01}$$

[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Unified MLLM: Harnessing Multi-Modalities

- Any-to-Any MLLM
  - NExT-GPT

➢ Key Aspect-II: Modality-switching Instruction Tuning



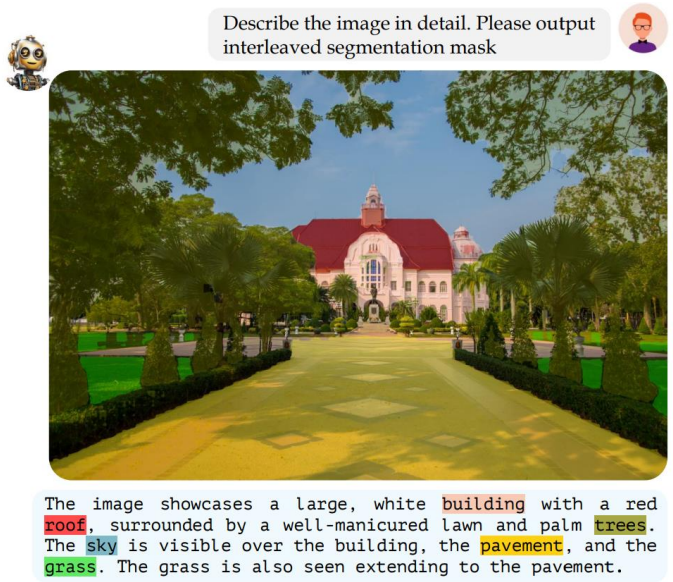
[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

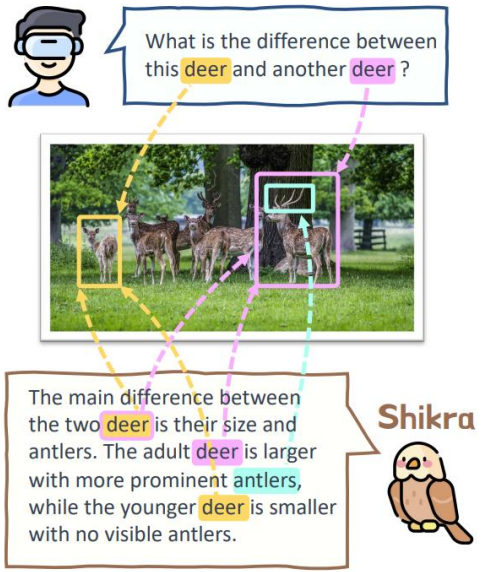
Fine-grained Capability of MLLM

Pixel-level Vision MLLM

*The vision MLLMs described above generally only support coarse-grained, instance-level visual understanding. This can lead to **imprecise visual interpretations**. Also due to the lack of visual grounding, these MLLMs will potentially **produce hallucinations**.*

- Visual Grounding
- Visual Segmentation
- Visual Editing
- Visual Inpainting
- ...

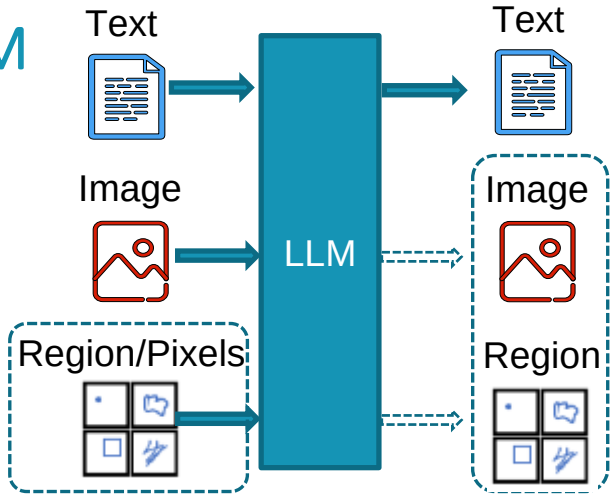




Fine-grained Capability of MLLM

- **Image-oriented Pixel-wise Regional MLLM**
  - GPT4RoI
  - NExT-Chat
  - MiniGPT-v2
  - Shikra
  - Kosmos-2
  - GLaMM
  - LISA
  - DetGPT
  - Osprey
  - PixelLM
  - LION
  - …

Users input an image (potentially specifying a region), and the LLM outputs content based on its understanding, grounding the visual content to specific pixel-level regions of the image.

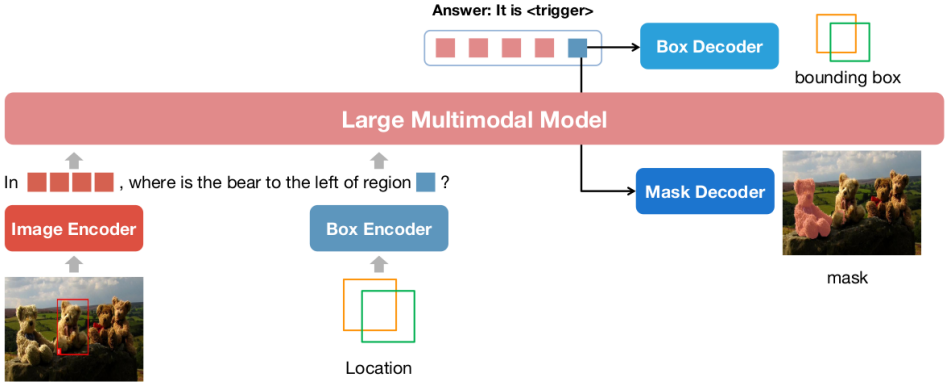


[1] GPT4RoI: Instruction Tuning Large Language Model on Region-of-Interest. 2023
[2] NExT-Chat: An LMM for Chat, Detection and Segmentation. 2023
[3] MiniGPT-v2: large language model as a unified interface for vision-language multi-task learning. 2023
[4] Osprey: Pixel Understanding with Visual Instruction Tuning. 2023
[5] GLaMM: Pixel Grounding Large Multimodal Model. 2023
[6] Kosmos-2: Grounding Multimodal Large Language Models to the World. 2023
[7] DetGPT: Detect What You Need via Reasoning. 2023
[8] PixelLM: Pixel Reasoning with Large Multimodal Model. 2023
[9] Lisa: Reasoning segmentation via large language model. 2023
[10] Shikra: Unleashing Multimodal LLM's Referential Dialogue Magic. 2023
…

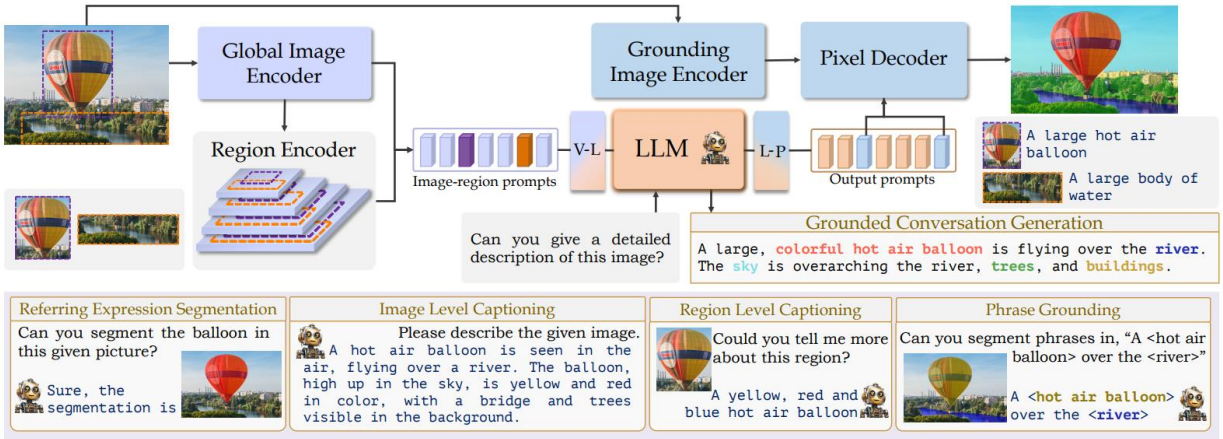
Fine-grained Capability of MLLM

- Image-oriented Pixel-wise

  - NExT-Chat

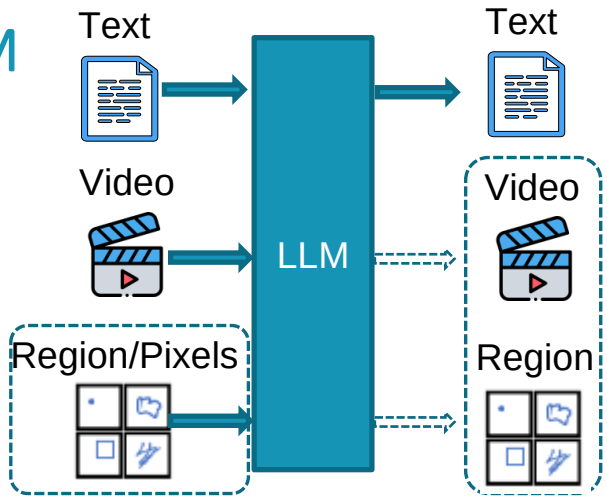


  - GLaMM



Fine-grained Capability of MLLM

- Video-oriented Pixel-wise Regional MLLM
  - PG-Video-LLaVA
  - Merlin
  - MotionEpic
  - …



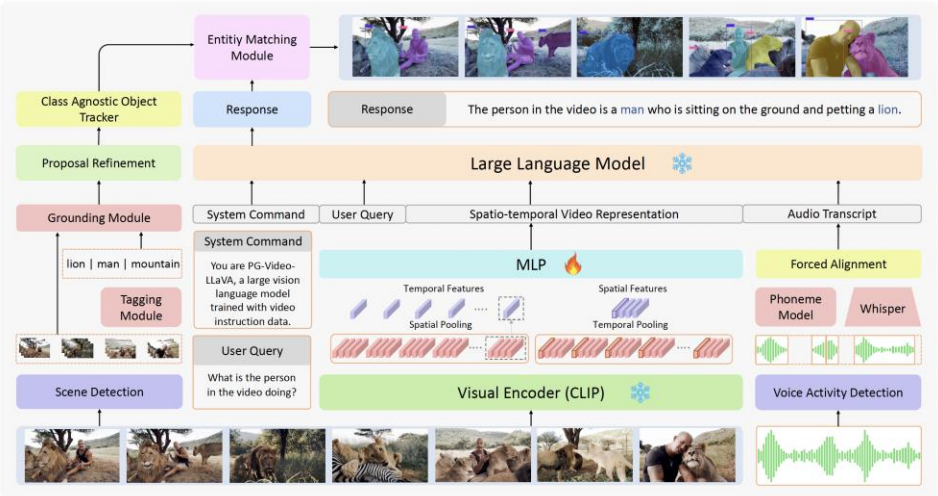
Users input an video (potentially specifying a region), and the LLM outputs content based on its understanding, grounding or tracking the content to specific pixel-level regions of the video.

[1] PG-Video-LLaVA: Pixel Grounding in Large Multimodal Video Models. 2023
[2] Merlin: Empowering Multimodal LLMs with Foresight Minds. 2023
[3] Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition. 2024
…

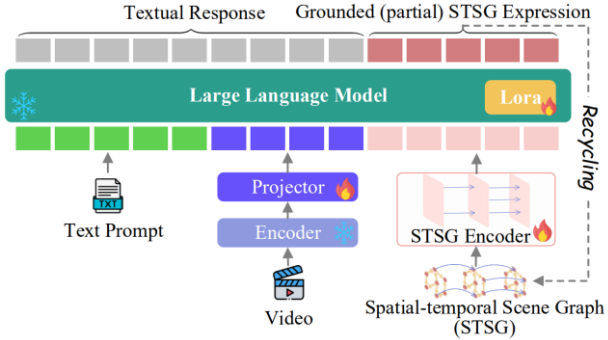
Fine-grained Capability of MLLM

- Video-oriented Pixel-wise Regional MLLM

PG-Video-LLaVA



MotionEpic





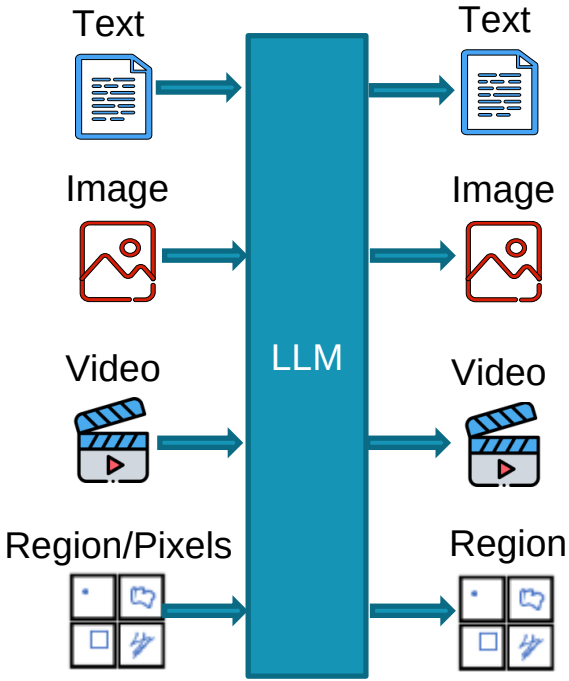
[1] PG-Video-LLaVA: Pixel Grounding in Large Multimodal Video Models. 2023
[2] Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition. 2024

Fine-grained Capability of MLLM

- Unified Pixel-wise MLLM
  - Vitron



Users input either an image or video (potentially specifying a region), and the LLM outputs content based on its understanding, generating, grounding or tracking the content to specific pixel-level regions of the image, video.



[1] VITRON: A Unified Pixel-level Vision LLM for Understanding, Generating, Segmenting, Editing. 2024

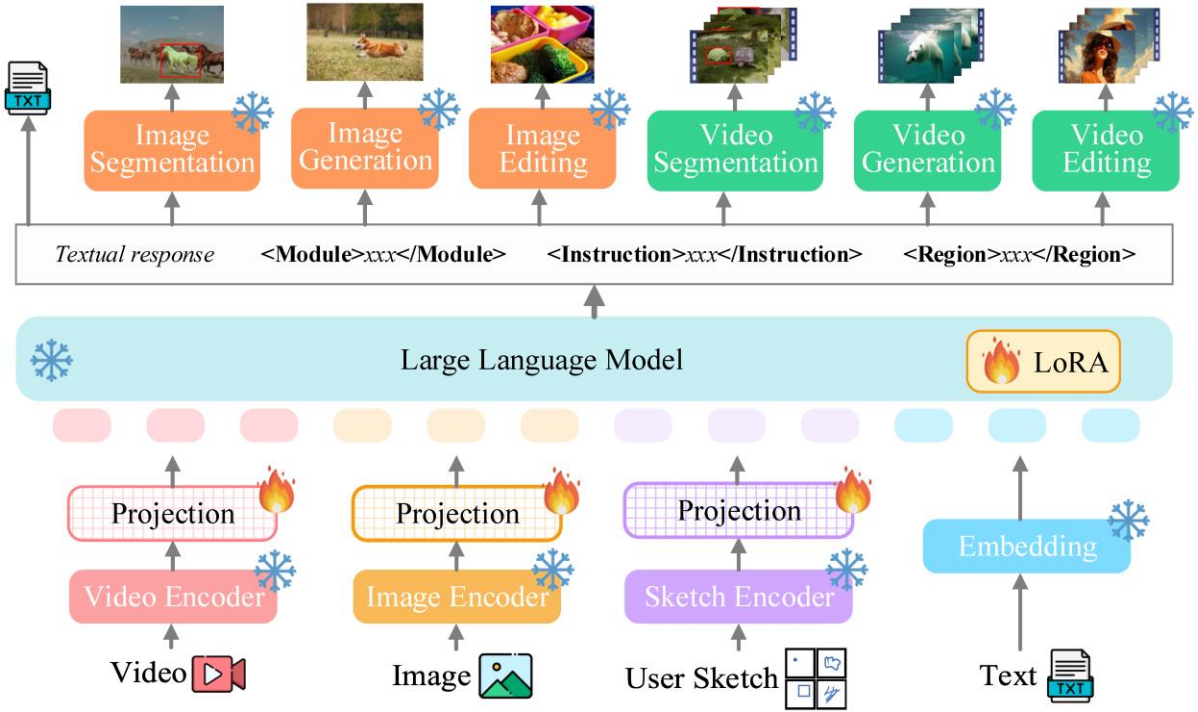
Fine-

- Unified
  - Vitron



Fine-grained Capability of MLLM

- Unified Pixel-wise MLLM
  - Vitron



[1] VITRON: A Unified Pixel-level Vision LLM for Understanding, Generating, Segmenting, Editing. 2024

2

Architecture of MLLM

How to design an MLLM?



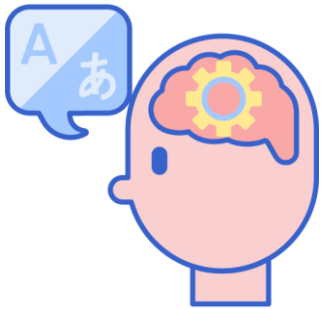
Overview of MLLM Architecture

- ## Preliminary Idea: Intelligence over Language

☞ *Due to the scaling law, **emergent phenomena** have extensively already occurred in language-based LLMs.*

☞ *These LLMs now generally possess very powerful **semantic understanding capabilities**.*

☞ *This also implies that **language is a crucial modality for carrying intelligence**.*



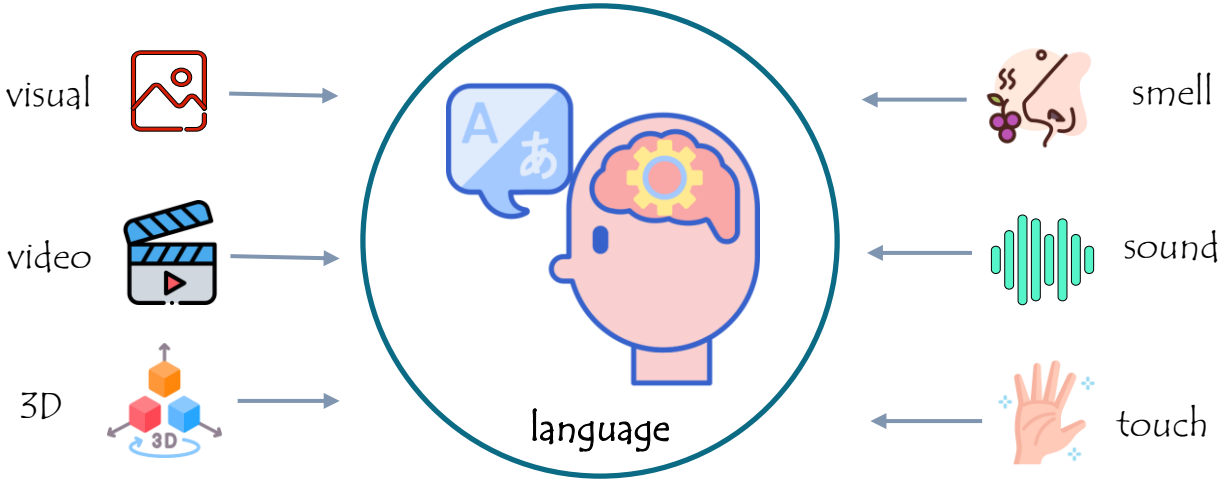
language

Overview of MLLM Architecture

- ## Preliminary Idea: Language Intelligence as Pivot

☞ *Given this premise, nearly all **CURRENT** MLLMs are built based on language-based LLMs as the core decision-making module (i.e., the brain or central processor).*

☞ *By adding additional external non-textual modality modules or encoders, LLMs are enabled with multimodal perceptual/operation abilities.*



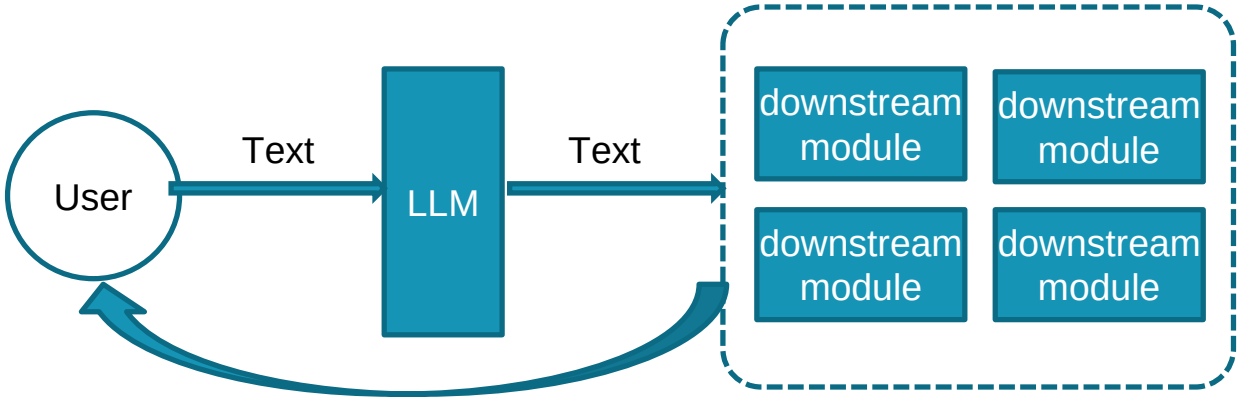
Overview of MLLM Architecture

Architecture-I: LLM as Discrete Scheduler/Controller

The role of the LLM is to receive textual signals and instruct textual commands to call downstream modules.

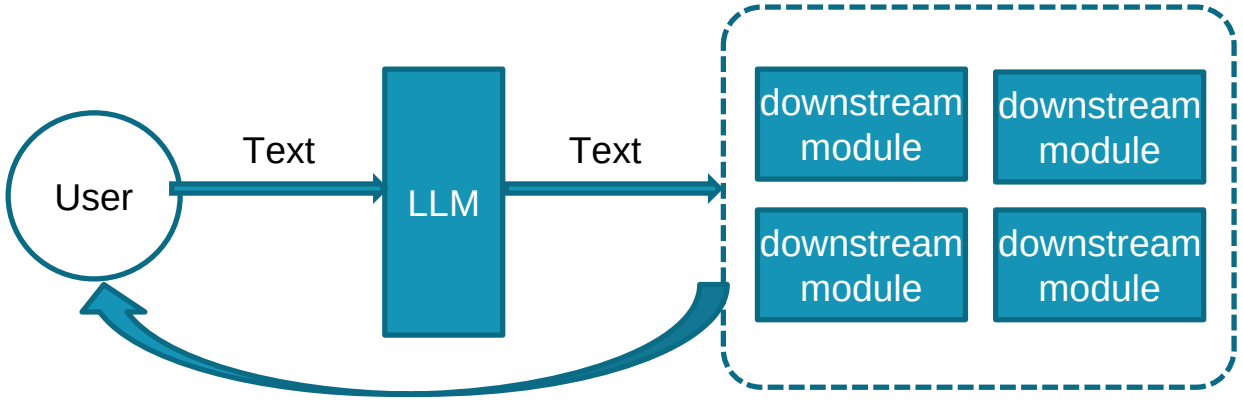
- Key feature:

  All message passing within the system, such as "multimodal encoder to the LLM" or "LLM to downstream modules", is facilitated through pure textual commands as the medium.



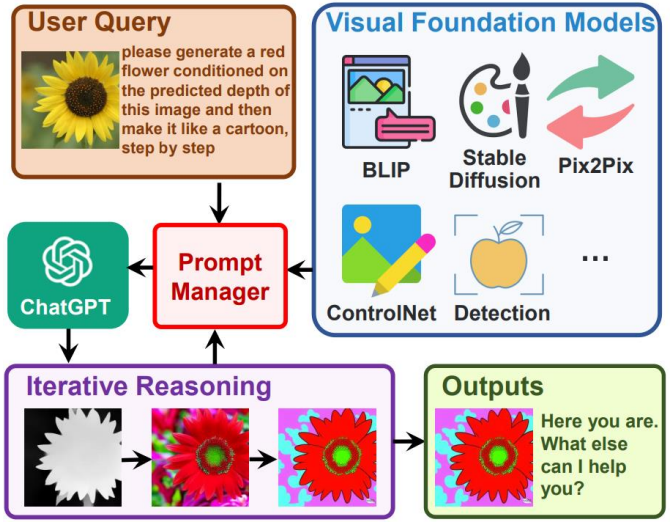
Overview of MLLM Architecture

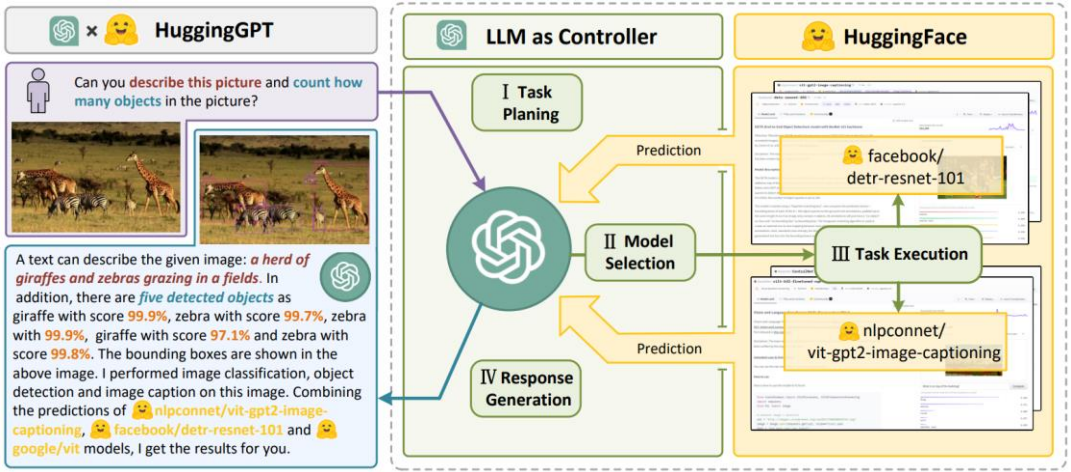
- **Architecture-I: LLM as Discrete Scheduler/Controller**
  - Representative MLLMs:
    - Visual-ChatGPT
    - HuggingGPT
    - MM-REACT
    - ViperGPT
    - AudioGPT
    - LLaVA-Plus
    - ...



Overview of MLLM Architecture

- Architecture-I: LLM as Discrete Scheduler/Controller
  - Visual-ChatGPT
  - HuggingGPT





[1] Visual ChatGPT: Talking, Drawing and Editing with Visual Foundation Models. 2023
[2] HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face. 2023

Overview of MLLM Architecture

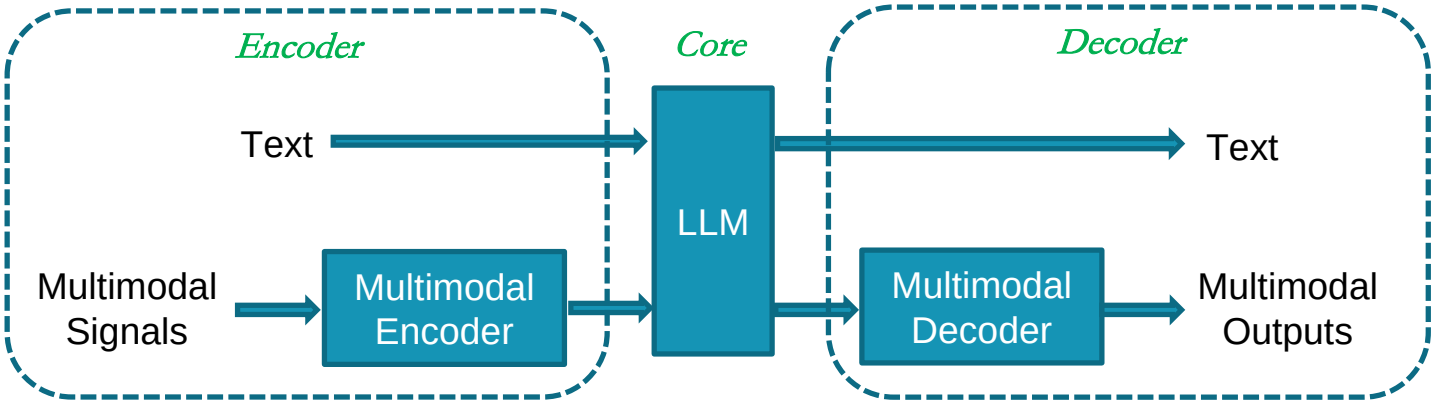
- ## Architecture-II: LLM as Joint Part of System



*The role of the LLM is to perceive multimodal information, and **react by itself**, in an structure of **Encoder-LLM-Decoder**.*

- Key feature:

  *LLM is the key joint part of the system, **receiving multimodal information directly from outside**, and delegating instruction to decoders/generators in a more smooth manner.*

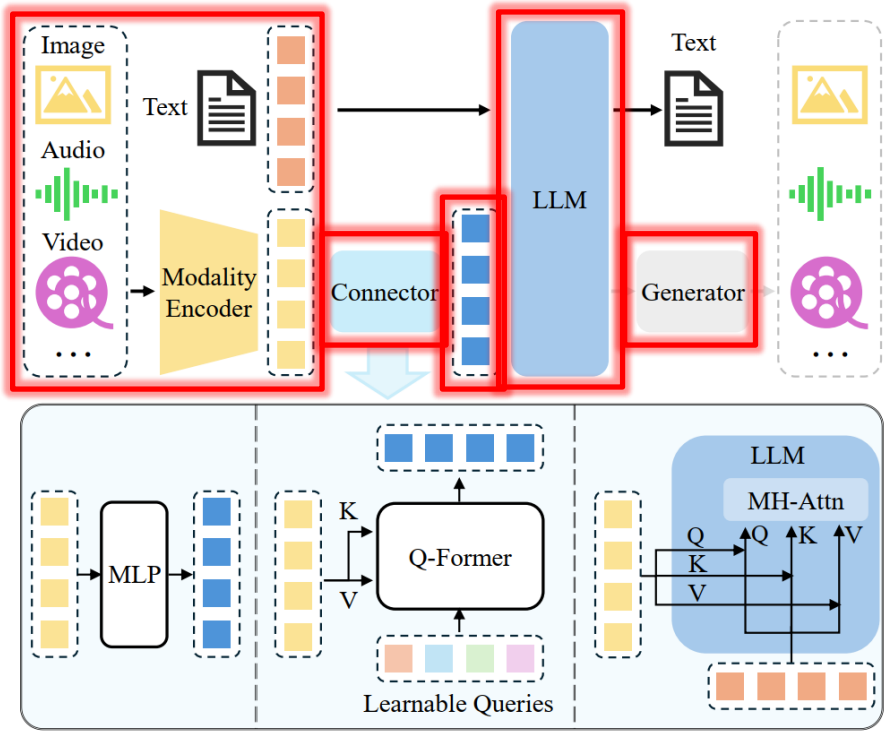


Overview of MLLM Architecture

- **Architecture-II: LLM as Joint Part of System**

  More promising

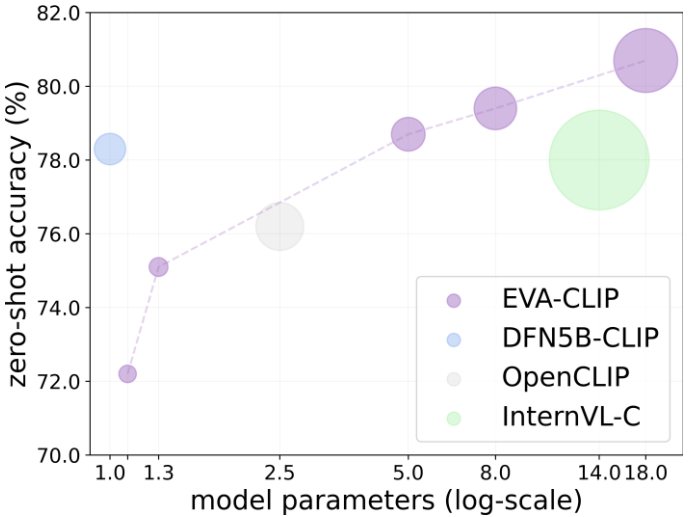
  - ≈ 96% MLLMs belong to this category.



[1] A Survey on Multimodal Large Language Models. https://github.com/BradyFU/Awesome-Multimodal-Large-Language-Models, *2023.*

Multimodal Encoding

- ## Visual (Image&Video) Encoder
  - **CLIP-ViT** is the most popular choice for vision-language models.
  - Advantages:
    - Providing image representations well aligned with text space.
    - Scale well with respect to parameters and data.

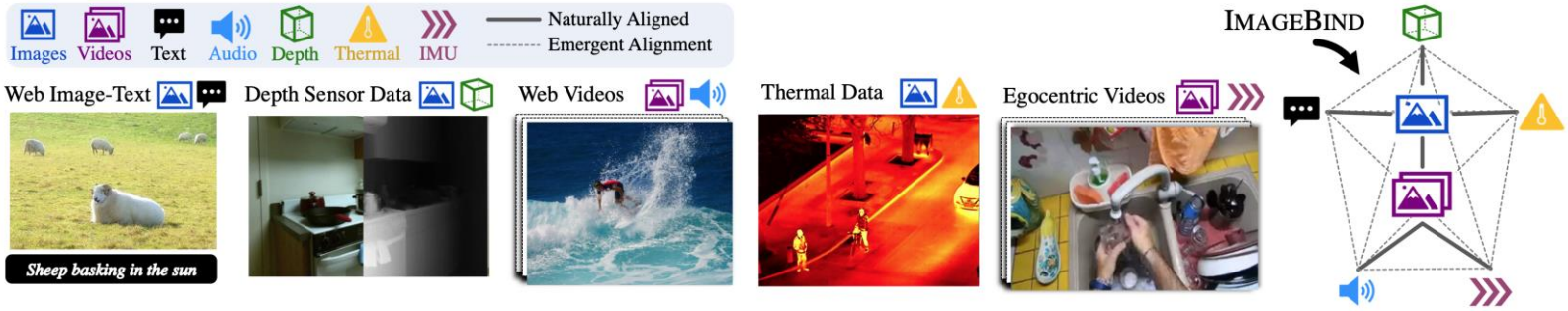


Multimodal Encoding

- Non-Visual Encoder
  - Audio:
    - HuBERT
    - Whisper
    - BEATs
  - 3D Point:
    - Point-BERT

Multimodal Encoding

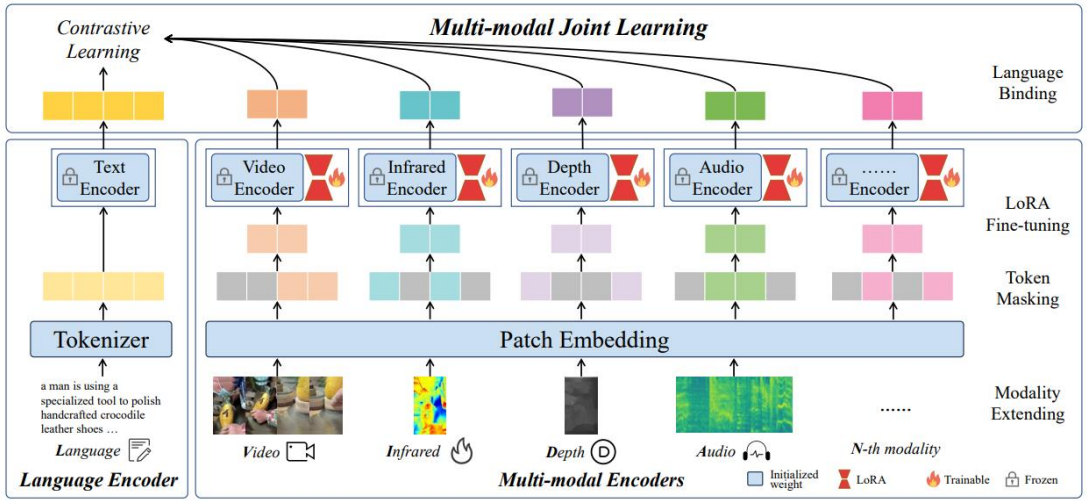
- ## Unified Multimodal Encoder
  - ImageBind:
    - Embedding all modalities into a joint representation space of Image.
    - Well aligned modality representations can benefit LLM understanding



[1] ImageBind: One Embedding Space To Bind Them All. 2023

Multimodal Encoding

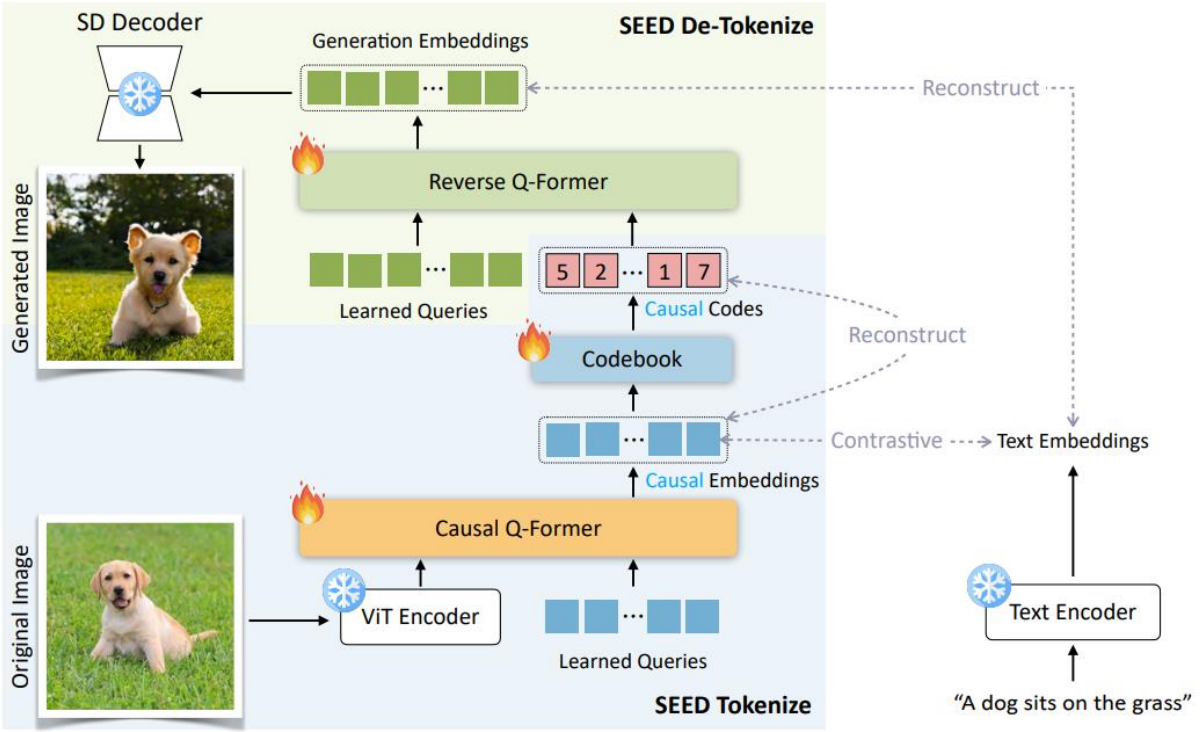
- ## Unified Multimodal Encoder
  - ### LanguageBind:
    - Embedding all modalities into a joint representation space of Language.
    - Well aligned modality representations can benefit LLM understanding



[1] LanguageBind: Extending Video-Language Pretraining to N-modality by Language-based Semantic Alignment. 2023

Multimodal Signal Tokenization

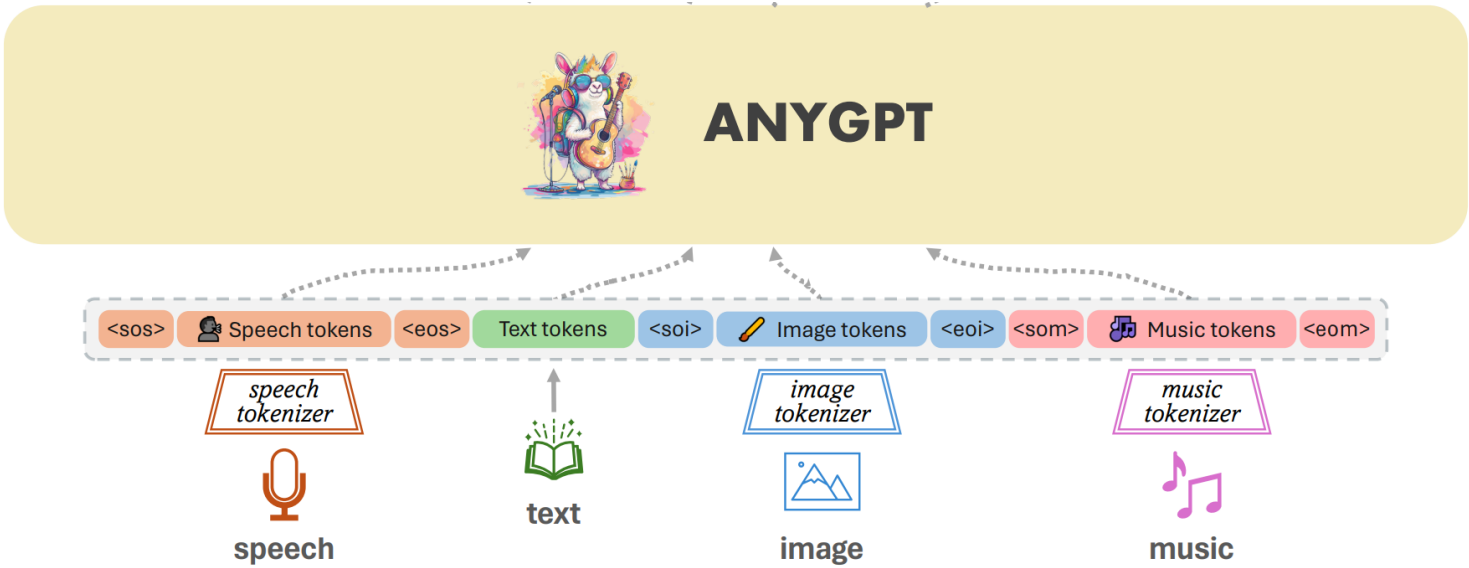
- Tokenization
  - SEED



[1] Planting a SEED of Vision in Large Language Model. 2023

Multimodal Signal Tokenization

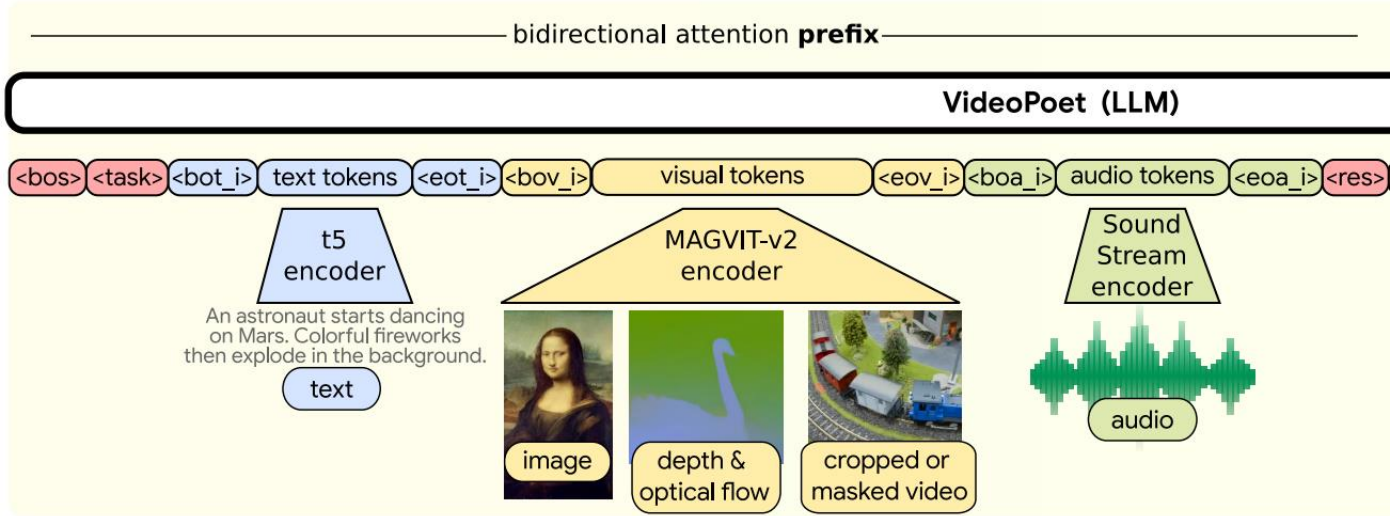
- Tokenization
  - AnyGPT



[1] AnyGPT: Unified Multimodal LLM with Discrete Sequence Modeling. 2023

Multimodal Signal Tokenization

- Tokenization
  - VideoPoet



[1] VideoPoet: A Large Language Model for Zero-Shot Video Generation. 2023

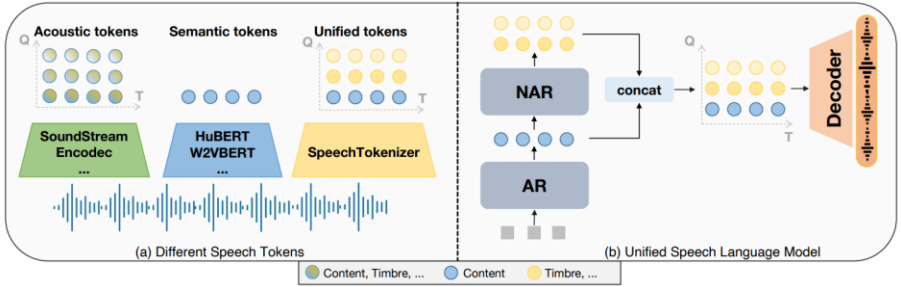
Multimodal Signal Tokenization

- ## Visual (Image&Video) Tokenization in Codebook
  - Represent multimodal signals as discrete tokens in a codebook
    - Advantages: support unified multimodal signal understanding and generation in an auto-regressive next-token prediction framework
    - More commonly used in image synthesize
      - Parti
      - Muse (parallel)
      - MaskGIT (parallel)
    - Representative Multimodal LLMs
      - Gemini
      - CM3
      - VideoPoet

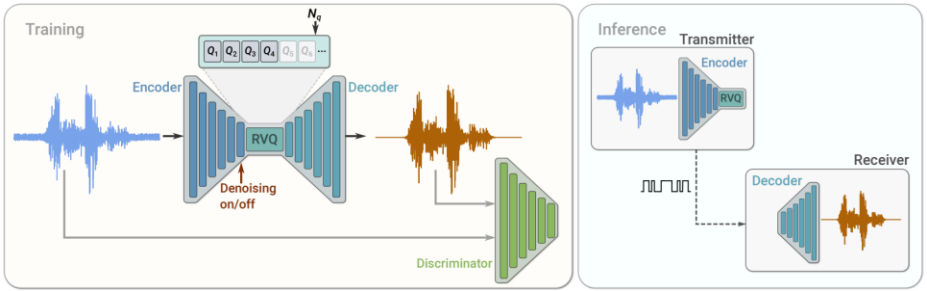
Multimodal Signal Tokenization

- ## Audio Tokenization

  - SpeechTokenizer +RVQ-VAE



  - SoundStream +RVQ-VAE

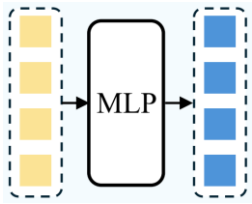


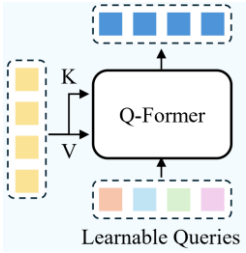
[1] SpeechTokenizer: Unified Speech Tokenizer for Speech Large Language Models. 2023
[2] SoundStream: An End-to-End Neural Audio Codec. 2021

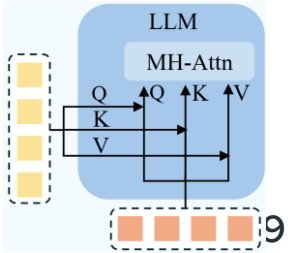
Input-side Projection

Methods to Connect Multimodal Representation with LLM

- Projecting multimodal (e.g., image) representations into LLM semantic space
  - Linear projection: **LLaVA, MiniGPT-4, NExT-GPT**
  - Two-layer MLP: **LLaVA-1.5/NeXT, CogVLM, DeepSeek-VL, Yi-VL**
  - Perceiver Resampler: **Flamingo, Qwen-VL, MiniCPM-V, LLaVA-UHD**
  - Q-Former: **BLIP-2, InstructBLIP, VisCPM, VisualGLM**
  - C-Abstractor: **HoneyBee, MM1**



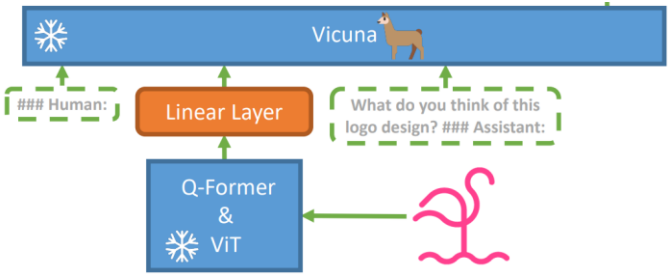




Input-side Projection

- ## Some Insights
  - Different papers have different conclusions about other projection methods.
  - Two-layer MLP is better than linear projection. (LLaVA)
  - Linear projection is more useful than Q-former layers. (MiniGPT-4)

	Method	LLM	Res.	GQA	MME	MM-Vet
	InstructBLIP	14B	224	49.5	1212.8	25.6
	Only using a subset of InstructBLIP training data					
0	**LLaVA**	7B	224	–	502.8	23.8
1	+VQA-v2	7B	224	47.0	1197.0	27.7
2	+Format prompt	7B	224	46.8	1323.8	26.3
3	+MLP VL connector	7B	224	47.3	1355.2	27.8
4	+OKVQA/OCR	7B	224	50.0	1377.6	29.6



Model	AOK-VQA	GQA
MiniGPT-4	58.2	32.2
(a) MiniGPT-4 w/o Q-Former	56.9	33.4
(b) MiniGPT-4 + 3 Layers	49.7	31.0
(c) MiniGPT-4 + Finetune Q-Former	52.1	28.0

[1] Improved Baselines with Visual Instruction Tuning. 2023
[2] MiniGPT-4: Enhancing Vision-Language Understanding with Advanced Large Language Models. 2021

Backbone LLMs

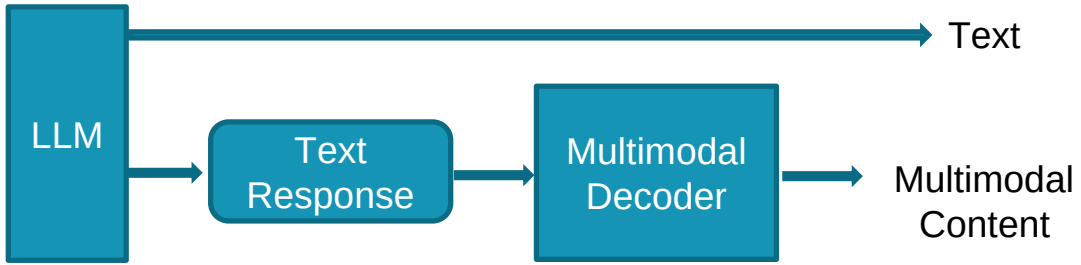
- ## Open-source Language-based LLMs

LLM	Size (B)	Data Scale (T)	Date	Language	Architecture
Flan-T5	3/11	-	Oct-2022	en, fr, de	Encoder-Decoder
LLaMA	7/13	1.4	Feb-2023	en	Decoder
Alpaca	7	-	Mar-2023	en	Decoder
Vicuna	7/13	1.4	Mar-2023	en	Decoder
LLaMA-2	7/13	2	Jul-2023	en	Decoder
GLM	2/10	0.4	Oct-2022	en	Decoder
Qwen	1.8/7/14	3	Sep-2023	en, zh	Decoder
Skywork	13	3.2	Oct-2023	en	Decoder

[1] A Survey of Large Language Models. https://github.com/RUCAIBox/LLMSurvey, 2023

Decoding-side Connection

- **Message passing via 1) discrete token of language**
  - Representative MLLMs:
    - Visual-ChatGPT
    - HuggingGPT
    - GPT4Video
    - MM-REACT
    - ViperGPT
    - ModaVerse
    - Vitron
    - …



  - Pros:
    - High performance lower-bound
    - More Efficient, i.e., without tuning
  - Cons:
    - Loss of end-to-end tuning capabilities.
    - Performance upper-bound is limited, i.e., some multimodal signals cannot be optimally conveyed through text).

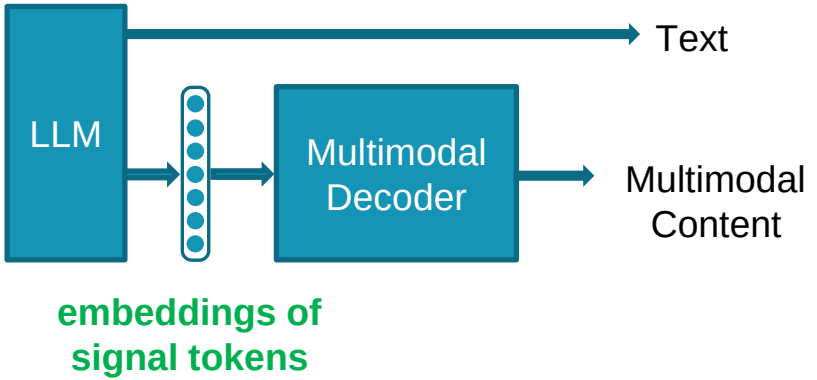
[1] Visual-ChatGPT: Talking, Drawing and Editing with Visual Foundation Models. 2023
[2] HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face. 2023
[3] ModaVerse: Efficiently Transforming Modalities with LLMs. 2024
[4] VITRON: A Unified Pixel-level Vision LLM for Understanding, Generating, Segmenting, Editing. 2024

Decoding-side Connection

- **Message passing via 2) continuous embedding**

  *Passing the message from LLM to downstream decoders via soft embeddings, i.e., **signal tokens**.*

  - Merits
    - Capable of end-to-end tuning, resulting in more efficient instruction transmission
    - More able to convey various multimodal signals that text alone cannot express, e.g.,
      - *the numeration of vision*
      - *the visual-spatial relational semantics*



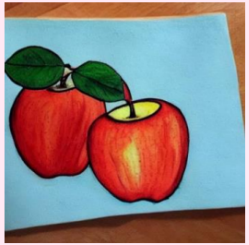
[1] Generating Images with Multimodal Language Models. 2023
[2] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Decoding-side Connection

- Message passing via 1) discrete token of language
  - ***Instruction***: *There were originally 7 apples on the table, but I ate one. Then, how many apples are left now? Please generate a picture to describe the result.*

Response: Sure, I'd be happy to help you understand the process and results.
Based on your statement, "There were originally 7 apples on the table, but I ate one", I inferred that there are now 6 apples left on the table.
To visualize this, I used a text-to-image model to generate an image based on your description. The model I used is called 'runwayml/stable-diffusion-v1-5', which is currently the only available model for this task. The image generated can be found at the following path: '/images/ec4d.jpg'.
Please note that the image is a representation of the text and may not be 100% accurate. If there's anything else you need, feel free to ask!

HuggingGPT



{***Intermedia Caption:*** 6 apples left on the table}

[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Decoding-side Connection

- Message passing via 2) continuous embedding
  - ***Instruction***: *There were originally 7 apples on the table, but I ate one. Then, how many apples are left now? Please generate a picture to describe the result.*

Response: Certainly! In the end, you have the following outcomes: 6 apples on the table.

NExT-GPT



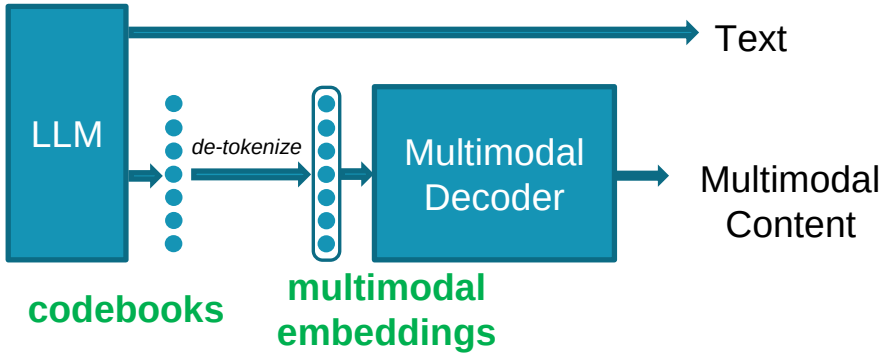
[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Decoding-side Connection

- **Message passing via 3) codebooks**

  *LLM generates special tokens id, i.e., **codebooks**, to downstream (visual) decoders .*

  - Merits
    - Capable of end-to-end tuning for higher efficiency in command transmission
    - Better at expressing various multimodal signals that cannot be captured by text alone
    - Supports autoregressive multimodal token generation



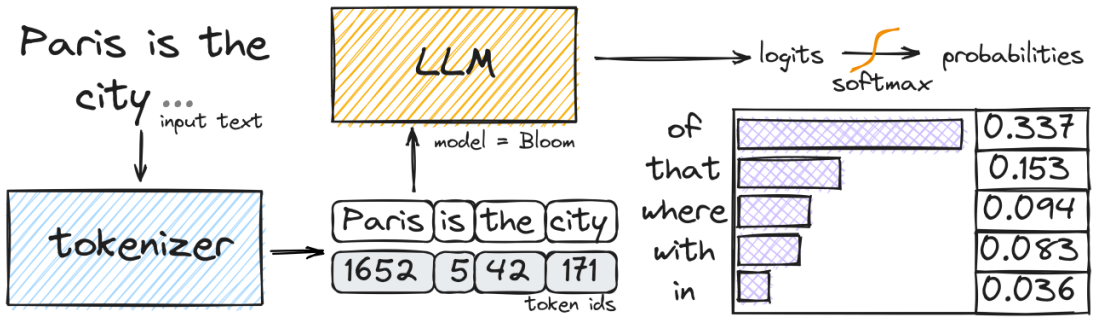
[1] Unified-IO 2: Scaling Autoregressive Multimodal Models with Vision, Language, Audio, and Action. 2023
[2] LVM: Sequential Modeling Enables Scalable Learning for Large Vision Models. 2023
[3] AnyGPT: Unified Multimodal LLM with Discrete Sequence Modeling. 2024
[4] VideoPoet: A Large Language Model for Zero-Shot Video Generation. 2024

Multimodal Generation

- ## Text Generation

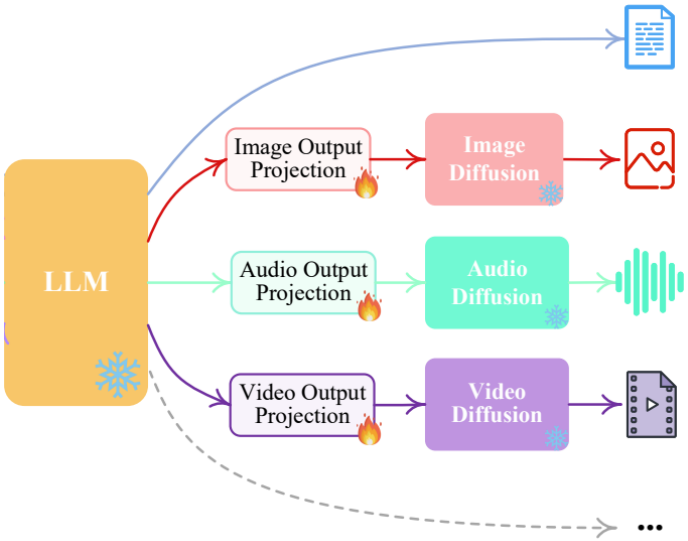
  - LLMs naturally support direct text generation

    via e.g., BPE decoding, Beam search, ...



Multimodal Generation

- **Generation via Diffusion Models**
  - Visual (Image/Video) Generator
    - Image Diffusion
    - Video Diffusion
  - Audio Generator
    - Speech Diffusion
    - Audio Diffusion

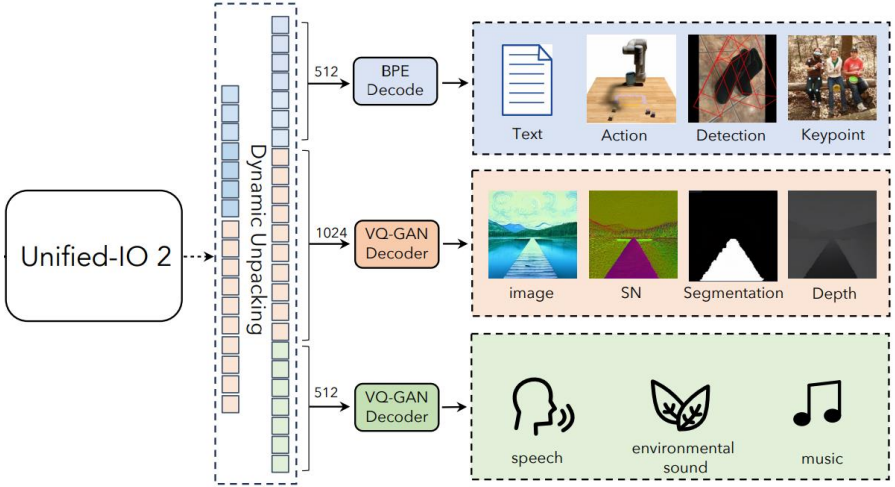


[1] NExT-GPT: Any-to-Any Multimodal LLM. 2023

Multimodal Generation

- ## Generation via Codebooks

  - ### Visual (Image/Video) Generator
    - VQ-VAE + Codebooks
    - VQ-GAN + Codebooks
  - ### Audio Generator
    - SpeechTokenizer + Residual Vector Quantizer
    - SoundStream + Residual Vector Quantizer



[1] Unified-IO 2: Scaling Autoregressive Multimodal Models with Vision, Language, Audio, and Action. 2023

Multimodal Generation

- ## Generation via Codebooks
  - ### VQ-GAN in Stable-diffusion
    - $64 \times 64 \times 3$ or $32 \times 32 \times 4$

Encoder	Decoder
$x \in \mathbb{R}^{H \times W \times C}$	$z_{\mathbf{q}} \in \mathbb{R}^{h \times w \times n_z}$
$\text{Conv2D} \to \mathbb{R}^{H \times W \times C'}$	$\text{Conv2D} \to \mathbb{R}^{h \times w \times C''}$
$m \times \{\text{Residual Block, Downsample Block}\} \to \mathbb{R}^{h \times w \times C''}$	$\text{Residual Block} \to \mathbb{R}^{h \times w \times C''}$
$\text{Residual Block} \to \mathbb{R}^{h \times w \times C''}$	$\text{Non-Local Block} \to \mathbb{R}^{h \times w \times C''}$
$\text{Non-Local Block} \to \mathbb{R}^{h \times w \times C''}$	$\text{Residual Block} \to \mathbb{R}^{h \times w \times C''}$
$\text{Residual Block} \to \mathbb{R}^{h \times w \times C''}$	$m \times \{\text{Residual Block, Upsample Block}\} \to \mathbb{R}^{H \times W \times C'}$
$\text{GroupNorm, Swish, Conv2D} \to \mathbb{R}^{h \times w \times n_z}$	$\text{GroupNorm, Swish, Conv2D} \to \mathbb{R}^{H \times W \times C}$

Table 7. High-level architecture of the encoder and decoder of our *VQGAN*. The design of the networks follows the architecture presented in [25] with no skip-connections. For the discriminator, we use a patch-based model as in [28]. Note that $h = \frac{H}{2^m}$, $w = \frac{W}{2^m}$ and $f = 2^m$.

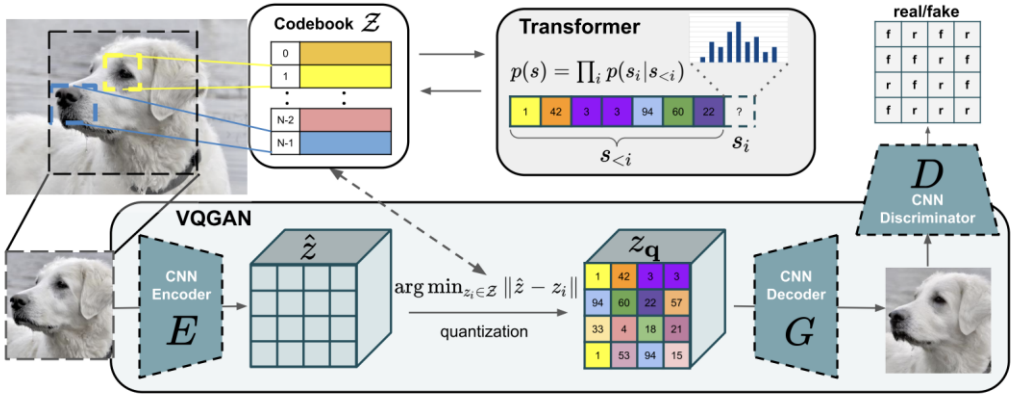


Figure 2. Our approach uses a convolutional *VQGAN* to learn a codebook of context-rich visual parts, whose composition is subsequently modeled with an autoregressive transformer architecture. A discrete codebook provides the interface between these architectures and a patch-based discriminator enables strong compression while retaining high perceptual quality. This method introduces the efficiency of convolutional approaches to transformer based high resolution image synthesis.

Model	Stage-1 (latent space learning)	Latent Space	Stage-2 (prior learning)
VQ-VAE	VQ-VAE	Discrete (after quantization)	Autoregressive PixelCNN
VQGAN	VQGAN (VQ-VAE + GAN + Perceptual Loss)	Discrete (after quantization)	Autoregressive GPT-2 (Transformer)
VQ-Diffusion	VQ-VAE	Discrete (after quantization)	Discrete Diffusion
Latent Diffusion (VQ-reg)	VAE or VQGAN	Continuous (before quantization)	Continuous Diffusion

Thanks!

Any questions?

